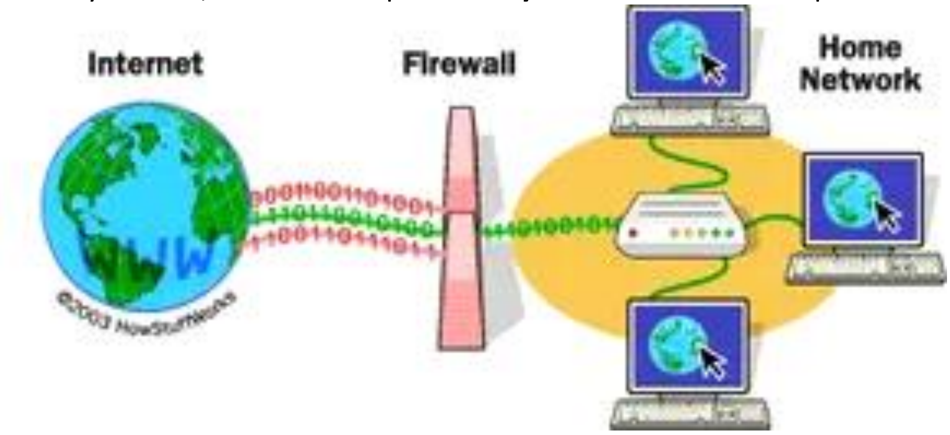
What is Firewall?

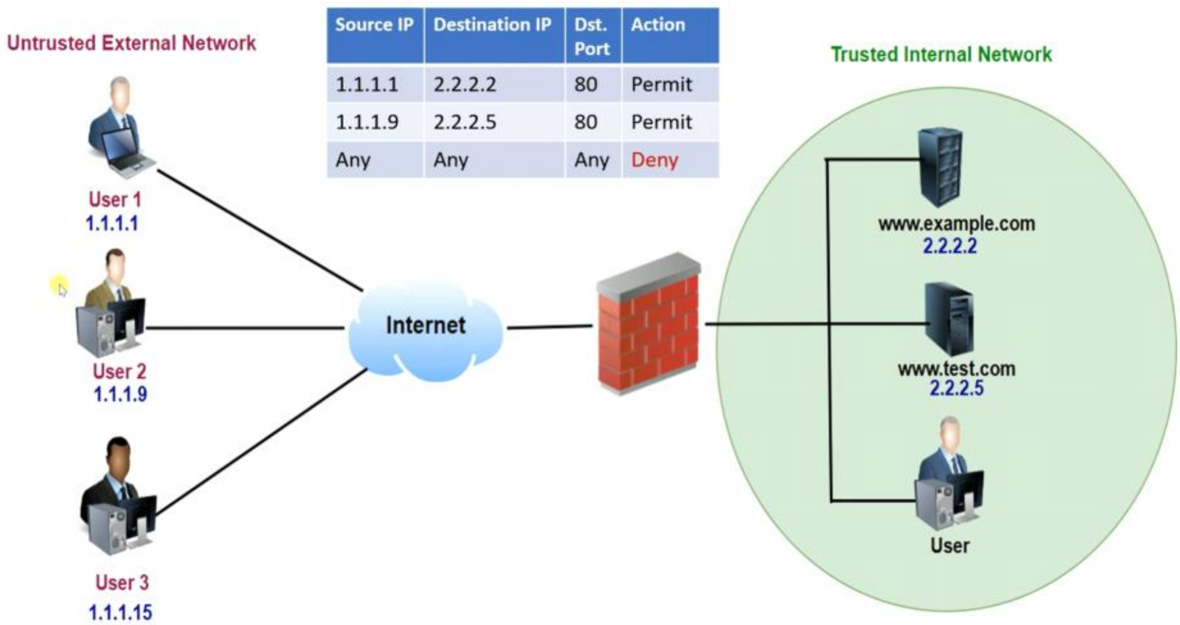
- It's the 1st line of defense.
- All incoming and outgoing network traffic will be passing only through it.
- It establishes a barrier between the trusted internal network(s) and the untrusted outside networks.
- It monitors all the incoming and outgoing network traffic that is passing through it and decides whether to permit or deny any specific traffic based on the set of security policy defined in it.
- By default, there's an implicit "**deny**" rule for unmatched packet.



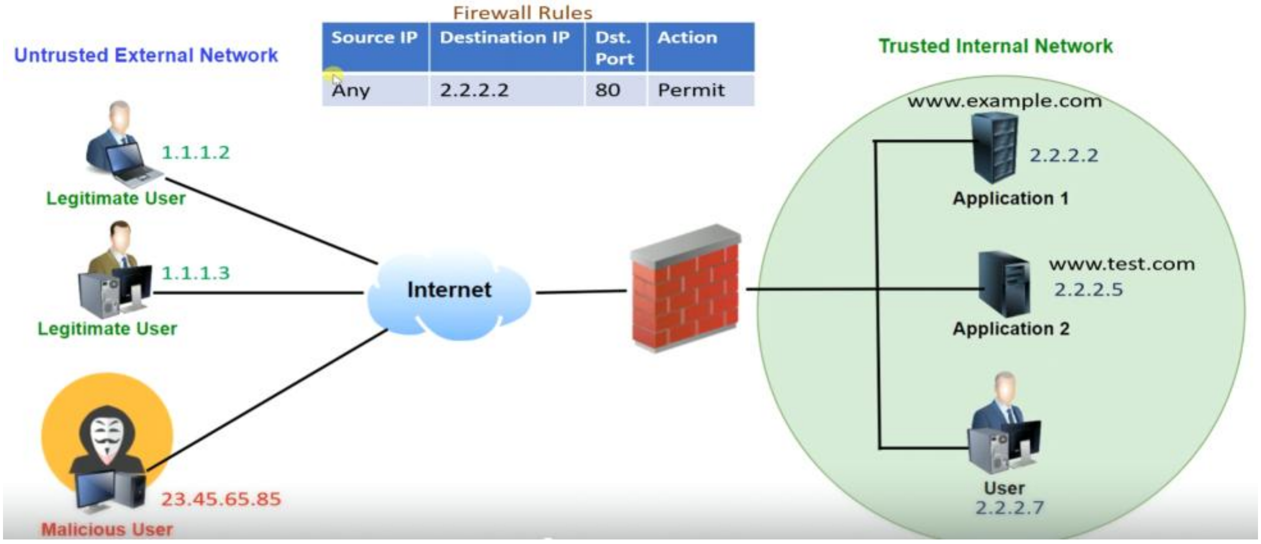
Simple legacy firewall rule:

Source IP	Destination IP	Dst. Port	Action
1.1.1.1	2.2.2.2	80	Permit
1.1.1.9	2.2.2.5	80	Permit
Any	Any	Any	Deny

Basic legacy network:



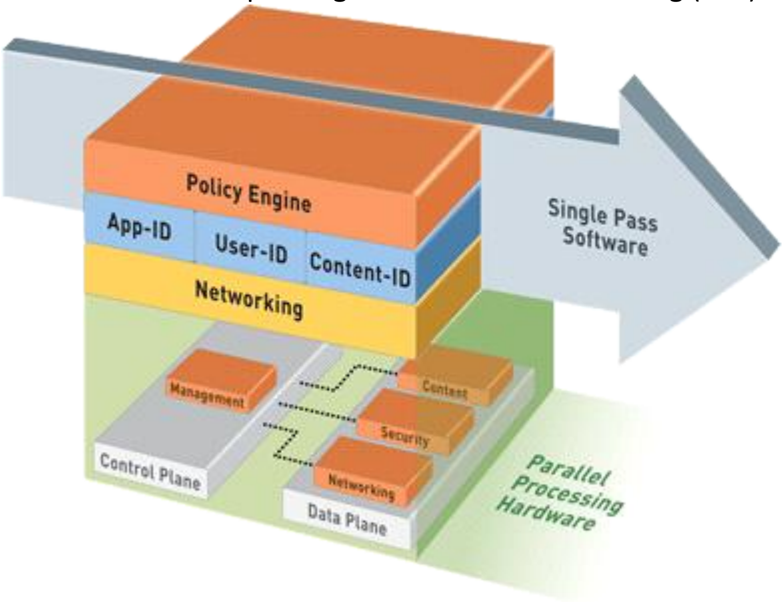
Note - Issue with legacy firewall is that it only works on L3 & L4 TCP/IP layers which checks only the source and the destination IP address. This could give access unauthorized access to the malicious user/attacker/hacker.



In this with legitimate users, Malicious user will also get the access to the www.example.com as source IP is allowed as "any" which gives the access to unwanted users too.

Palo-Alto Firewall

- Palo-Alto is a name of a city in California's San Francisco Bay Area in USA.
- PA was developed by Nir Zuk.
- Palo-Alto Networks (company) has a global footprint presence in 50+ countries, 24/7 support.
- It's a Next-Generation Firewall (NGFW) with advance features like:
    - Deep packet inspection
    - Intrusion detection and Intrusion Prevention system
    - Distributed Denial of Service (DDOS) protection.
    - Application-level inspection
    - User identification
    - Web content filtering,
        - Etc.
- It helps to identify, control and inspect the SSL encrypted traffic and applications.
- PA has a real-time content scanning feature to protect against viruses and malwares.
- It helps to protect against the spyware, data leakage and application vulnerabilities.
- It can help to prevent certain functions of an application without blocking the entire application.
- PA is based on unique Single Pass Parallel Processing (SP3) architecture.



- SP3 architecture provides high throughput and low latency network security service.
- SP3 has 2 components:
    - **Single Pass Software**
        - It performs the operation once per packet.
        - When a packet comes in, it performs all the operation at a same time in a single pass at once like:
            - networking function,
            - policy lookup,
            - application identification,
            - signature validation against malware or threats
            - content identification,
            - user identification and so on.
    - **Parallel Processing Hardware**

        - The parallel processing hardware ensures the single pass software works smoothly and efficiently.
        - It has separate control plane and data plane with multiple cores and processors for smooth and better performance.

- 3 main advance features of PA firewall:
    - **App-ID** - identify the application
    - **Content-ID** - Scan the content
    - **User-ID** - Identify the user

Palo-Alto VM using Azure

Thursday, December 25, 2025 8:19 PM

Minimum requirements:

- **16 GB RAM**
- **4 vCPUs**
- **224 GB HDD**/SSD
- VM Image: Palo-Alto (**Panaroma**)
- Username: <given at the time of VM creation>
- Password: <given at the time of VM creation>
- IP Address: <given at the time of VM creation>
- URL to access PA mgmt portal: https://<IP-Address>

Deployment using Azure (link):

1. Palo Alto Networks Professional Service Flex Licensing Migration Lab

1.1 Overview

The Following Lab guide will help you to understand how to migrate Non-Flex licensed Software Firewalls from Non-Flex license model (ELA, etc) to the new Flex License Model. It will also cover how to create Deployment profiles in the Customer Suport Portal (CSP) to cover several secanrios. The Lab will only cover Migration use cases as listed below

1. Standalone Firewall with Access to the CSP
2. Standalone Firewall, No Access to the CSP
3. Panorama-Managed Firewalls with Access to the CSP
4. Panorama-Managed Firewalls, No Access to the CSP

Private Cloud and other Public Cloud Providers will not be covered in the Lab. The deployed firewall are running in PanOS 9.1, 10.0.3, 10.2

1.2 Covered Secaniros

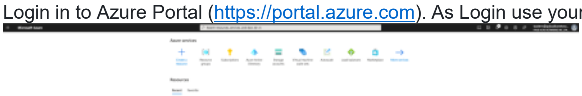
The following Secanrios and Lab activies are covered

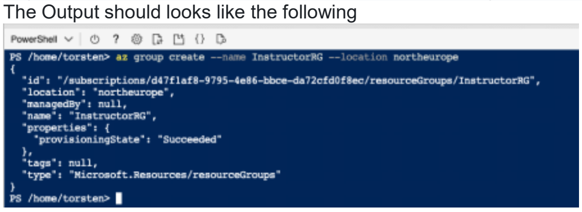
1. Deploy a new Lab Panorama to fullfill the Migration process
2. Configure Panorama to perform the Lab activities
3. Setup the Customer support Portal (CSP)
    1. Creating several Deployment Profiles
4. Deploy Software Firewalls and License them with an ELA License
    1. 2 Firewalls in PanOS 9.1.13-h3
    2. 2 Firewalls in PanOS 10.0.9
    3. 2 Firewalls in PanOS 10.2.3
5. Migrate Software Firewalls from NON-Flex License Model to Flex-License Model
    1. NON-Flex to Flex-License (Fixed Deployment Profile)
    2. Flex-License to Flex-License (Flexible Deployment Profile)
6. How to update the Deployment Profile
    1. Enable/Disable CDSS
    2. Increase/Decrease vCPU count
7. Troubleshooting

2 Deploy Panorama in Microsoft Azure

For this workshop you will create a first a Panorama before we deploy the Software Firewall in other Public Cloud Providers. The Panorama will have direct internet access. the Panorama is not connected to any other internal Ressource in Azure.

2.1 Deploy Azure Resource Group

1. Login in to Azure Portal (https://portal.azure.com). As Login use your Palo Alto Networks Credentials

2. Open Azure Cloud Shell

3. In Cloud Shell execute the following command but change before the values [StudentRGName] and [Location] Available Regions are: North Europe, East US, UK South, UAE North, Australia Central
az group create --name [StudentRGName] --location [Location] --tags Owner=Workshop-DeleteMe
4. The Output should looks like the following


2.2 Deploy Panorama in Azure

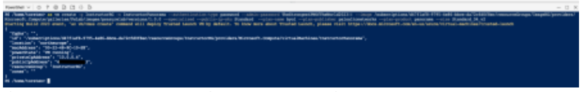
As next we will create the Panorama from a pre-staged image, after successfully creating the Resource Group.

1. Please go back to the Azure Cloud Shell
2. In the following command updat the following variables with yours:
    1. [StudentRGName] #Use the same same of the previous created Resource Group in the Chapter [Deploy new Resource Group in Azure]
    2. [VM-Name]
    3. [YourPassword]

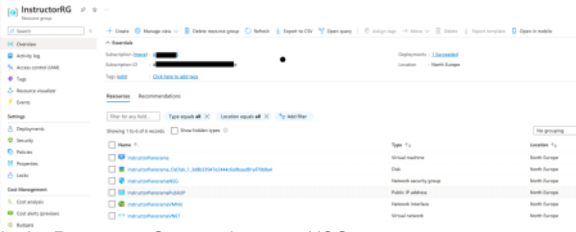
Don't change any other variables

```
az vm create -g [StudentRGName] -n [VM-Name] --authentication-type password --admin-password [YourPassword] --image /subscriptions/d47f1af8-9795-4e86-bbce-da72cfd0f8ec/resourceGroups/ImageRG/providers/Microsoft.Compute/galleries/PsLab/images/psazurelab/versions/1.0.0 --specialized --public-ip-sku Standard  --plan-name byol --plan-publisher paloaltonetworks --plan-product panorama --size Standard_D4_v2
```

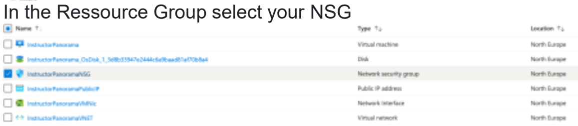
1. After you made the changes, execute the command in Azure Cloud Shell
2. The Output should looks like the following



3. Check your Ressource Group in Aure if the Deployment is completed

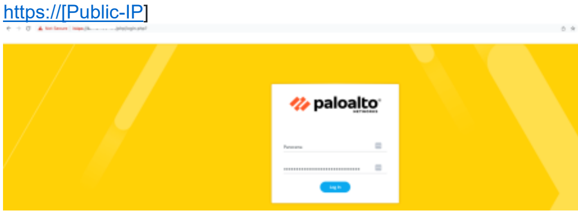


4. In the Ressource Group select your NSG



5. Now create an Inbound Security Rule to allow any traffic to your newly created Panorama

6. Login to your Panorama via the Public IP associated to it. The Instructor will provide you the Usernam and Password.
    1. [https://[Public-IP]]



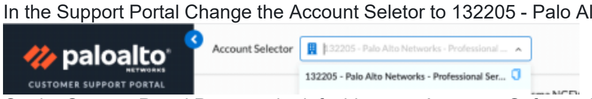
Congratulations!! You have succesfully deploye a Panorama in Microsoft Azure.

3. Customer Support Portal

In the following Lab section we will go to the Customer support portal (CSP) to create your first Deployment Profiles. This is needed for the intial Migration and generating a Serialnumber for the Panorama

3.1 Login To Customer Support Portal

1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



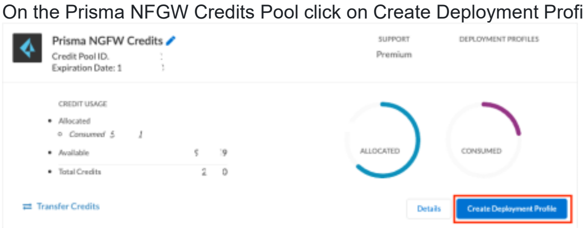
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits

3.2 Create Fixed Deployment Profiles

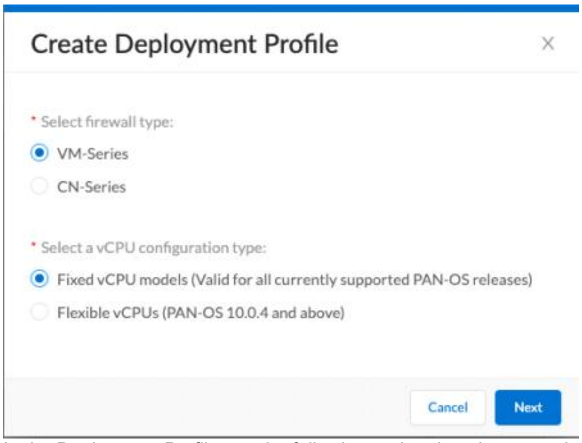
Now you will create one Deployment Profile in the Customer Support Portal.

3.2.1 Azure Deployment Profile

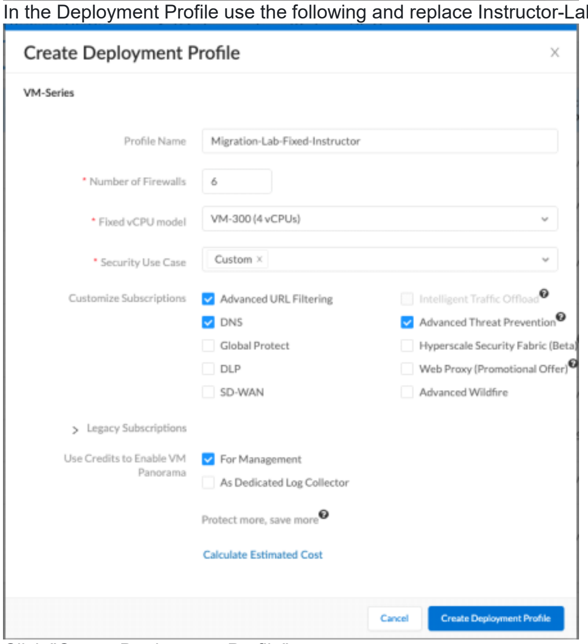
1. On the Prisma NFGW Credits Pool click on Create Deployment Profile



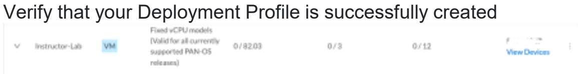
2. Select the following as shown on the picture below and click Next



3. In the Deployment Profile use the following and replace Instructor-Lab under "Profile Name" with "Migration-Lab-Fixed-[StudentName]"



4. Click "Create Deployment Profile"
5. Verify that your Deployment Profile is successfully created

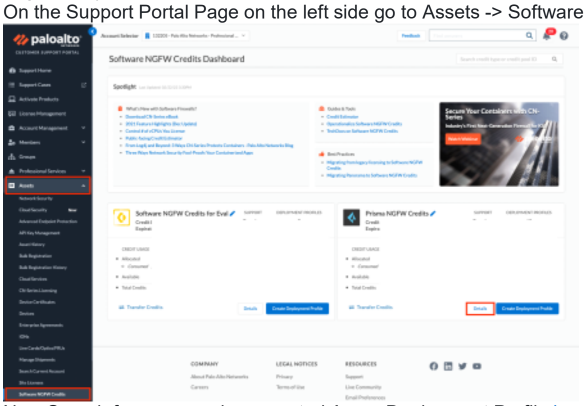


3.3 License Panorama

In the next steps you will create a Serialnumber for your previous created Panorama with the Flex License Credits

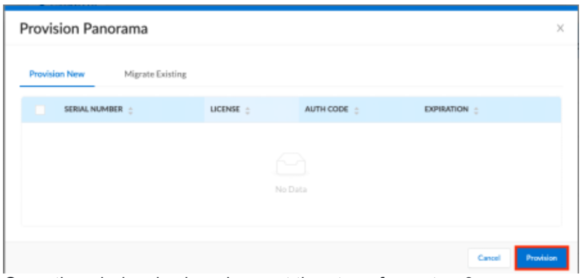
3.3.1 Provision Panorama Serialnumber

1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits -> Details



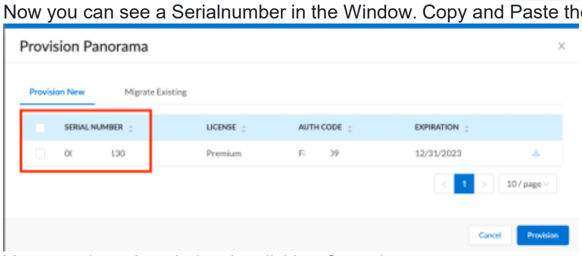
3. Now Search for your previous created Azure Deployment Profile here
4. Click on the 3 dots and on Provision Panorama

5. In the new window click on Provision



6. Once the window is closed repeat the steps from step 3

7. Now you can see a Serialnumber in the Window. Copy and Paste the Serialnumber



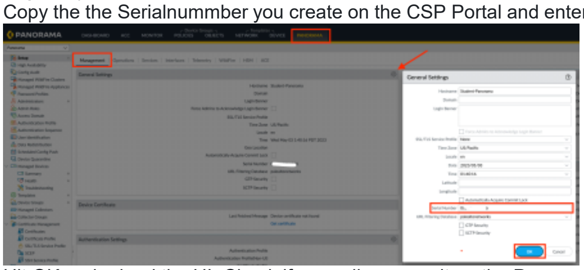
8. You can close the window by clicking Cancel

3.3.2 Configure Panorama

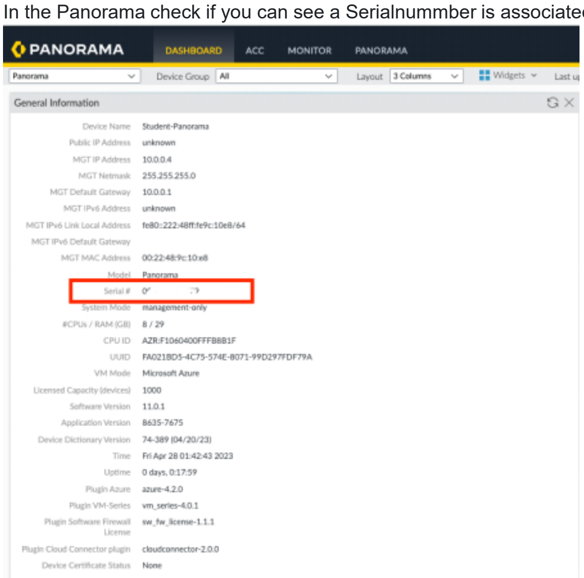
As next we will License your Panorama with the Serialnumber you created above and create a new Decive Group and Template inside your new Panorama and do some basic configuration in your Device Template

3.3.2.1 License Panorama

1. Login to your Panorama [https://[Public-IP]](https://[Public-IP])
2. Copy the the Serialnummber you create on the CSP Portal and enter it under the Panorama Tab -> Setup -> Management -> General Settings

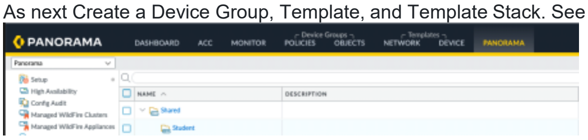


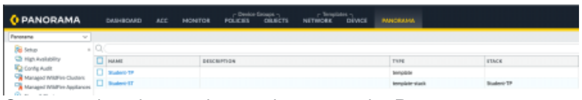
3. Hit OK and reload the UI. Check if a pending commit on the Panorama is needed. If yes, commit to Panorama.
4. In the Panorama check if you can see a Serialnummber is associated to it



3.3.2.2 Create Device Group and Device Template

1. As next Create a Device Group, Template, and Template Stack. See the picture below as example

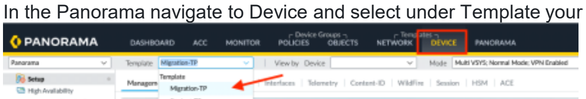




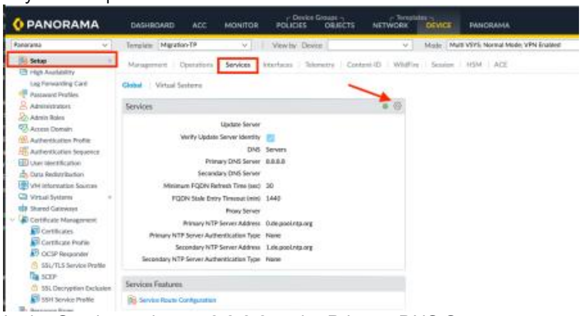
2. Once you done it commit your changes to the Panorama

3.3.2.3 Base config of the Device Template

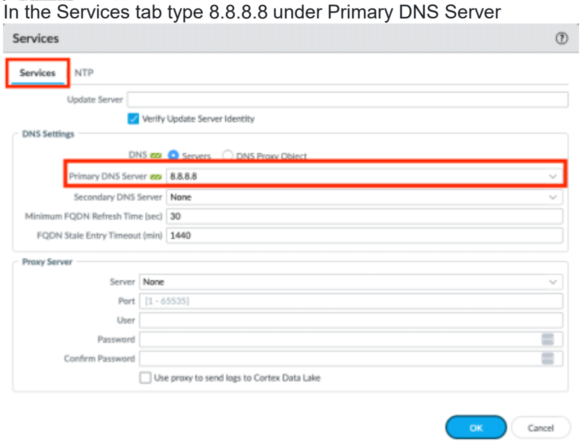
1. In the Panorama navigate to Device and select under Template your previous create Template (my example Stundent-TP)



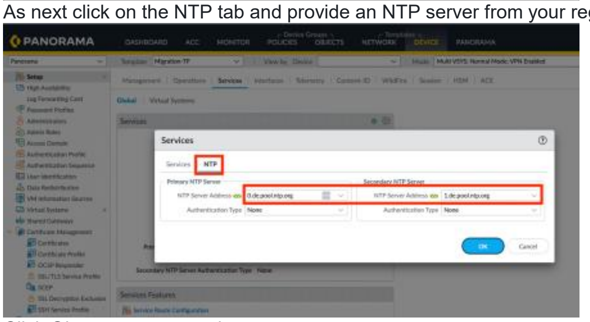
2. In your Template click on Select -> Service and click on the wheel.



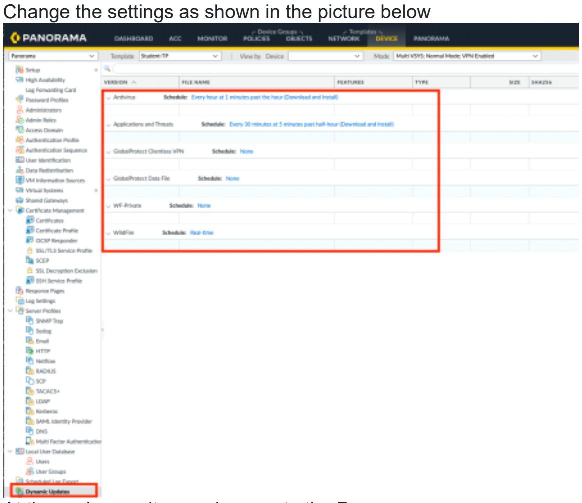
3. In the Services tab type 8.8.8.8 under Primary DNS Server



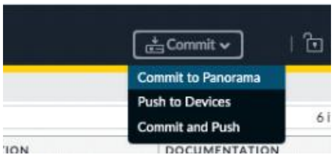
4. As next click on the NTP tab and provide an NTP server from your region (my example 0.de.pool.ntp.org)



5. Click Ok once you entert it
6. As next click on the left panel on Dynamic Updates
7. Change the settings as shown in the picture below



8. At the end commit your changes to the Panorama



4. Deploy Firewalls in Azure

In the following chapter you will deploy several Software Firewalls in different PanOS version. The Software Firewalls will automatically join your previous created Panorama

4.1 What you will do?

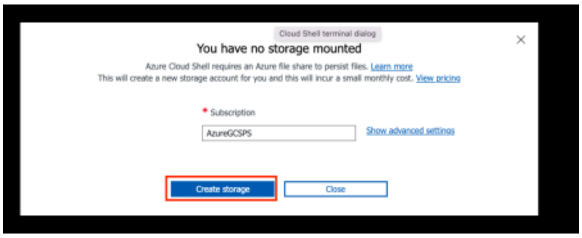
- Login to Azure Portal (https://portal.azure.com) and login with your Credentials
- Download Terraform Code from GitHub
- Modify Terraform Code
- Execute Terraform Code
- Validate Deployment in Azure Portal and Panorama

4.2 Deployment

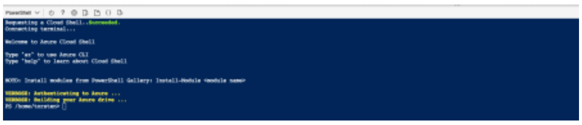
1. Login in to Azure Portal (https://portal.azure.com)

2. Open Azure Cloud Shell

3. click on Create storage. In some case it will not create a Storage Account. In that case click in "Show advanced settings" and create your own storage account.

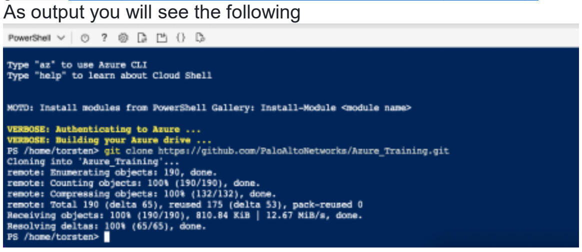


4. Once the creation of the storage is completed you will see the following



5. Download Terraform Code from GitHub

   1. in the Cloud shell execute the following command
      git clone https://github.com/PaloAltoNetworks/flex-license-migration-lab.git
   2. As output you will see the following



6. Now browse to the deployment folder folder
   cd cd flex-license-migration-lab/azure/single\ firewall\ deployment/

7. Rename the terraform.tfvars.example to terraform.tfvars mv ./example.tfvars terraform.tfvars
   Expand For Details
   Command: mv ./terraform.tfvars.example terraform.tfvars

8. Modify the ```terraform.tfvars``` inside Cloud shell with the ```vi``` command 1. Modify the following variables in the File.
resource_group_name = "migration-[Studenname]" #replace [Studentname] with your Name
password = "SecurePassWord12!!" #change the password. Use a complex password
storage_account_name = "pantfstorage[Studenname]" #replace [Studentname] with your name in small letters without space
storage_share_name = "bootstrapshare[Studenname]" #replace [Studentname] with your name in small letters without space

```
location                 = "north europe"
resource_group_name      = "migration-test2"
common_vmseries_sku      = "byol"
vm_series_version_set1   = "9.1.13"
vm_series_version_set2   = "10.0.9"
vm_series_version_set3   = "10.2.3"
username                 = "panadmin"
password                 = "SecurePassWord12!!"
allow_inbound_mgmt_ips = [
  "0.0.0.0/0", # OR add your own public IP address here, visit "https://ifconfig.me/"
]
firewall1 = "Firewall1"
firewall2 = "Firewall2"
firewall3 = "Firewall3"
firewall4 = "Firewall4"
firewall5 = "Firewall5"
firewall6 = "Firewall6"

storage_account_name    = "pantfstoragetest"
storage_share_name      = "bootstrapsharetest"

avzones = ["1", "2", "3"]

files = {
  "files/authcodes"    = "license/authcodes" # authcode is required only with common_vmseries_sku = "byol"
  "files/init-cfg.txt" = "config/init-cfg.txt"
}
```

1. Save your changes by pressing ESC and type :wq! and ENTER
2. As next switch to the folder files and rename the init-cfg.sample.txt to init-cfg.txt using the mv command
3. Modify the init-cfg.txt inside Cloud shell with the vi command. Make sure you added the same name of the Device Group and Template Stack you created in your Panorama. The value for the varibles tplname and dgname can be found in the section 3.3.2 License Panorama / Create Device Group and Template

```
type=dhcp-client
vm-auth-key=123456789012345   #Follow the fLink below to create/show the key
panorama-server=10.1.2.3      #change it to the Public IP of your Panorama
tplname=my-stack              #change it to the Template Stack inside your Panorama Section [3.3.2]
dgname=my-device-group        #change it to the Device Group inside your Panorama Section [3.3.2]
dhcp-send-hostname=yes
dhcp-send-client-id=yes
dhcp-accept-server-hostname=yes
dhcp-accept-server-domain=yes
```

LINK to Guide for the Key creation

1. As next in folder files and rename the authcodes.sample to authcodes using the mv command
2. Modify the authcodes files with the vi command.

```
XXXXXXX # Instructor will provide you the Key via Slack
```

1. Save your changes by pressing ESC and type :wq! and ENTER
2. Move back to the single\ firewall\ deployment folder with the command cd..
3. Once you made all your changes execute the Terraform code with following commands: 1. terraform init. 2. terraform plan. 3. terraform apply once you get the prompet type yes
4. Important! The complete deployment will take up to 10 Minutes after the completing the Terraform Apply. It is a good time for a break
5. Terraform Init

```
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset> terraform init
Initializing modules...
- inbound_bootstrap in ../modules/bootstrap
- inbound_lb in ../modules/loadbalancer
- inbound_scale_set in ../modules/vmss
- outbound_lb in ../modules/loadbalancer
- vnet in ../modules/vnet

Initializing the backend...

Initializing provider plugins...
- Finding hashicorp/azurerm versions matching "~> 2.26, ~> 2.42, 2.64.0, ~> 2.64"...
- Finding hashicorp/random versions matching "~> 3.0"...
- Installing hashicorp/azurerm v2.64.0...
- Installed hashicorp/azurerm v2.64.0 (signed by HashiCorp)
- Installing hashicorp/random v3.4.3...
- Installed hashicorp/random v3.4.3 (signed by HashiCorp)

Terraform has created a lock file .terraform.lock.hcl to record the provider
selections it made above. Include this file in your version control repository
so that Terraform can guarantee to make the same selections by default when
you run "terraform init" in the future.

Terraform has been successfully initialized!

You may now begin working with Terraform. Try running "terraform plan" to see
any changes that are required for your infrastructure. All Terraform commands
should now work.

If you ever set or change modules or backend configuration for Terraform,
rerun this command to reinitialize your working directory. If you forget, other
commands will detect it and remind you to do so if necessary.
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset>
```

6. Terraform Plan

```
Plan: 44 to add, 0 to change, 0 to destroy.

Changes to Outputs:
  + inbound_frontend_ips                = {
      + frontend01 = (known after apply)
    }
  + metrics_instrumentation_key_inbound = (sensitive value)
  + password                            = (sensitive value)
  + username                            = "panadmin"
```

7. Terraform Apply

```
Do you want to perform these actions?
  Terraform will perform the actions described above.
  Only 'yes' will be accepted to approve.

  Enter a value: yes
```

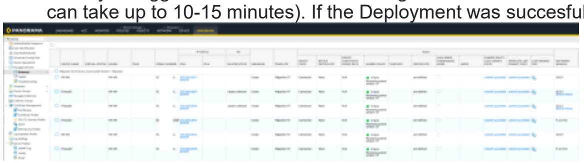
8. Once the terraform apply is completed you will see the following output

```
Apply complete! Resources: 44 added, 0 changed, 0 destroyed.

Outputs:

inbound_frontend_ips = {
  "frontend01" = "20.(        9"
}
metrics_instrumentation_key_inbound = <sensitive>
password = <sensitive>
username = "panadmin"
PS /home/torsten/Azure_Training/Azure_Autoscaling_Lab/vmseries_scaleset> 
```

4.3 Validate Deployment

- Login into Panorama
- Validate Deployment

1. Login into Panorama Public IP
2. Once you logged into the Panorama Navigate to the Panorama tab validate you can see your newly deployed Firewalls (The deployment and bootstrapping process can take up to 10-15 minutes). If the Deployment was succesful you will see the following output in Panorama -> Managed Devices -> Summary



1. You succesfull deployed your Environment if you can see the above output

Congratulations you succesfully deployed several VM-Series Firewalls in different PanOS Version and bootstrapped them.

5 License Migration

In the following steps you will migrate the previous created from a NON-Flex License model to the Flex-License model. You will do several migrations and create/update some Deployment profiles to fulfill the activities.

5.1 Covered Secanrios in Detail

1. Migrate all Software Firewall to Flex License model (Fixed Deployment Profile) via Panorama
2. Migrate one (1) Firewall with PanOS 9.1.13-h3 to Fixed License via Panorama
3. Migrate one (1) Firewall with PanOS 10.0.9 to Fixed License via Panorama
4. Migrate one (1) Firewall with PanOS 10.0.9 from Fixed License to Flex via Panorama
5. Migrate one (1) Firewall with PanOS 10.2.3 to Fixed License via Panorama
6. Migrate one (1) Firewall with PanOS 10.2.3 from Fixed License to Flex via Panorama
7. Migrate one (1) Firewall with PanOS 10.2.3 from Flex to Flex with increasing the vCPU via Panorama

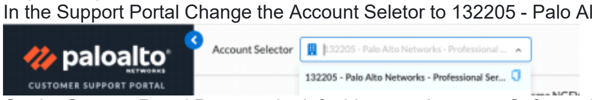
5.2 Migrate Software Firewalls

In the following section we will migrate now all Software Firewall from the NON-Flex license model to the Flex license model. For that we will use the Deployment profile fou created in the section [3.2 Create Fixed Deployment Profiles](#).

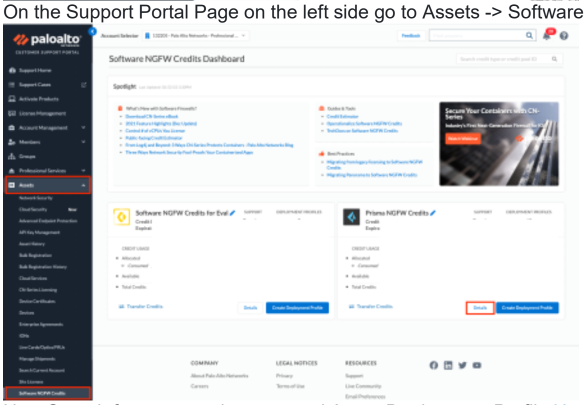
5.2.1 Initial Migration

In the following section you migrate all Software Firewalls from NON-Flex Licensing to Flex Licensing.

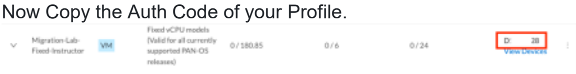
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



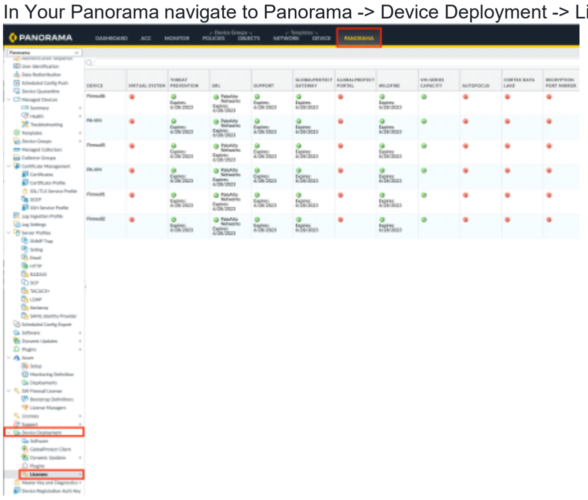
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits -> Details



4. Now Search for your previous created Azure Deployment Profile Here
5. Now Copy the Auth Code of your Profile.

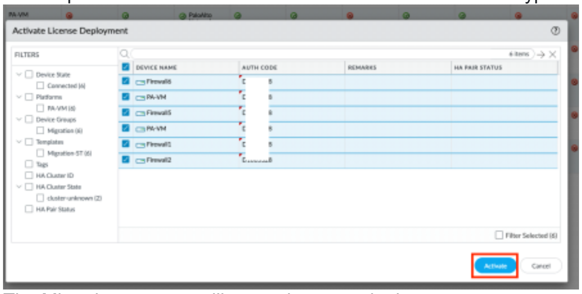


6. As next Login in to your Panorama https://[Public-IP]
7. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses

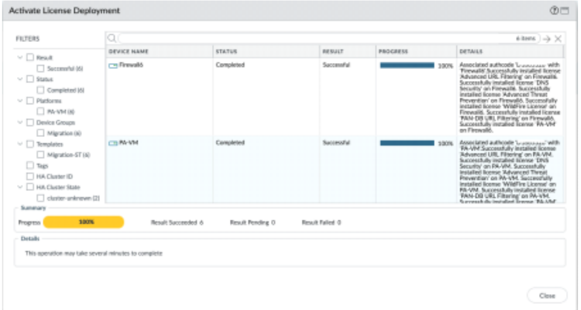


8. In the License window click at the bottom Activate

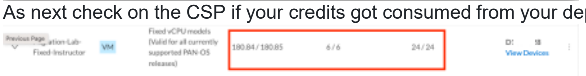
9. In the opened Window select now all available Firewalls and type in AUTH CODE field the auth code and click Activate



10. The Migration process will now take several minutes.
11. Once Migration is completed you will see the following outcome



12. As next check on the CSP if your credits got consumed from your deployment profile. You should see the below outcome

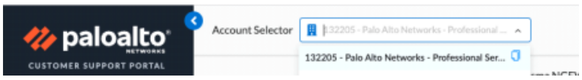


Congratulations!!! You Migrated successful all your Software Firewalls from a NON-Flex license model to Flex License model (Fixed Deployment Profile) via Panorama

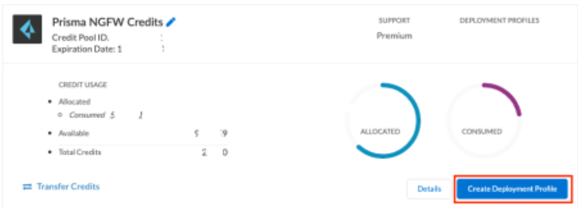
5.2.2 Migrate PanOS 10.0.9 to Flexible Deployment Profile

In the following section we will create a new Deployment Profile to migrate the Software Firewalls from a Fixed Deployment Profile to a Flexible Deployment Profile

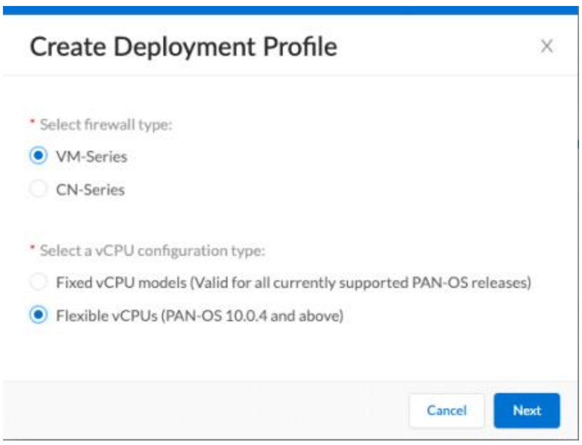
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



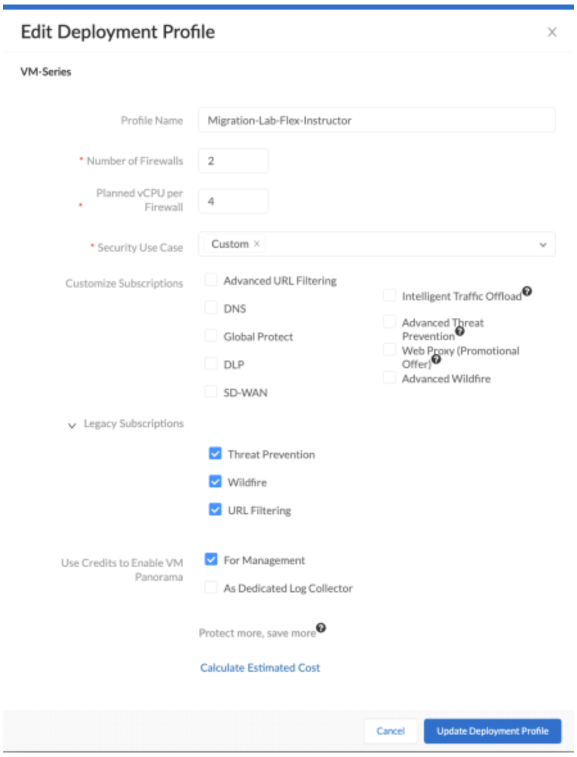
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma NFGW Credits Pool click on Create Deployment Profile



5. Select the following as shown on the picture below and click Next



6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-[StudentName]"



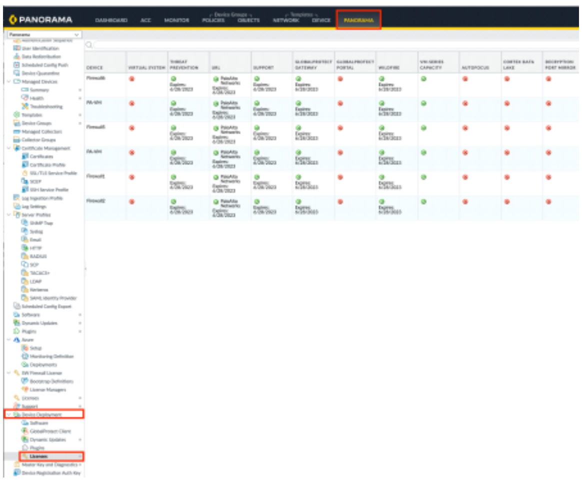
7. Click "Create Deployment Profile"

8. Verify that your Deployment Profile is successfully created

9. Now Copy the Auth Code of the newly created Deployment Profile

10. As next Login in to your Panorama [https://[Public-IP]](https://[Public-IP])

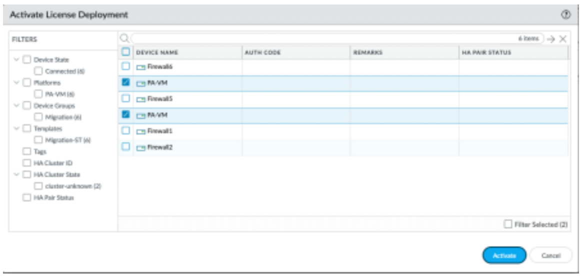
11. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses



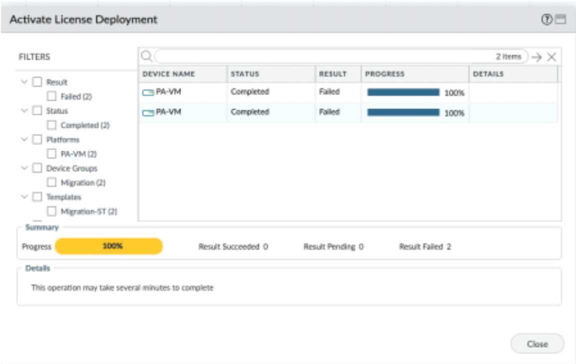
12. In the License window click at the bottom Activate

13. Select now Firewall 3 and 4 or as in shown in the Picture the firewalls with the name "PA-VM".

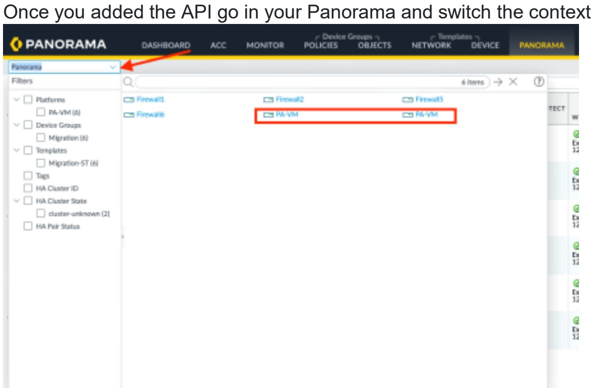
You can verify the Name of the firewalls in the Summary tab. Now type in AUTH CODE field the auth code and click Activate



14. Are the upgrade is working? If no, Why?
Expand For Details The Upgrade is not working because you chaning the VM capacity. You can only upgrade the Software Firewalls to the Flex Deployment profile when activating it directly on the VM and using the "Upgrade VM capacity"

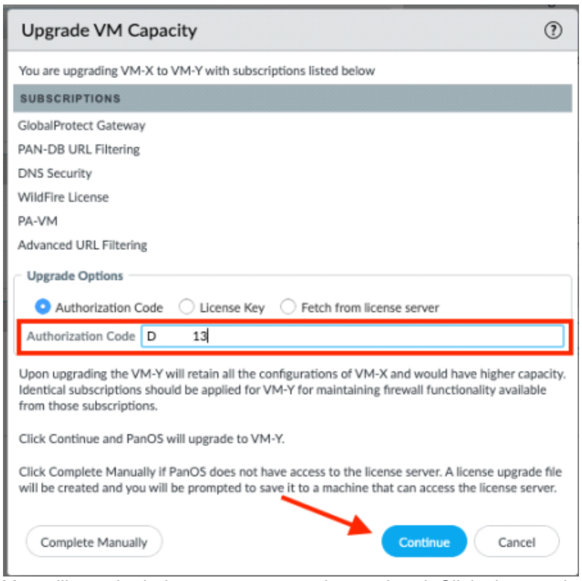


15. Before you can perform the License Key upgrade you have to install on the Software Firewalls the License API Key. Follow the [instructions] to perform the task. Repeat that
16. Install API License Key on ALL other Software Firewalls (1-6) too for future tasks
17. Once you added the API go in your Panorama and switch the context to Firewall 3 or 4 (or PA-VM)

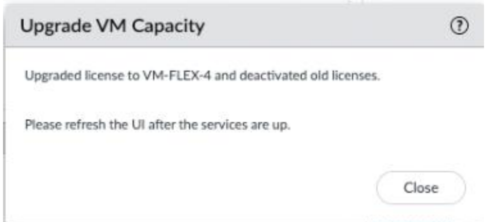


18. In the Firewall navigate Device -> License and click on Upgrade VM capacity

19. In the window add under Authorization Code your atuh code and click Continue



20. You will see the below outcome once it completed. Click close and refresh the UI



21. In the Firewall switch to the Dashboard and you can see the VM License changed to VM-FLEX-4

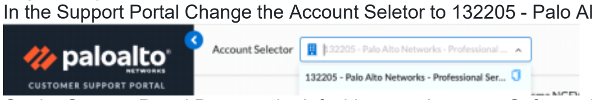
22. Repeat the same steps for the second firewall.
23. When you know go to the Support Portal and check your profiles, you can see that the count of the Fixed prile is reduced by 2 firewalls and 8 vcpus and the Flex profile increased.

Congratulations!!! You successful migrated 2 Firewalls from a Fixed License Deployment Profile to an Flexible Deployment profile and implemented the API License Key on the Firewalls

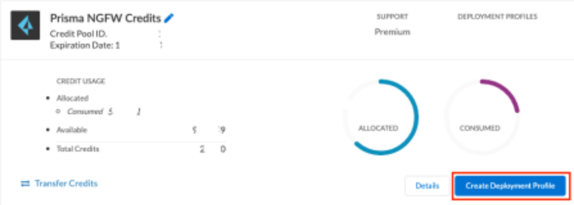
5.2.3 Migrate PanOS 10.2.3 to Flexible Deployment Profile

In the following section we will create a new Deployment Profile to migrate the Software Firewalls from a Fixed Deployment Profile to a Flexible Deployment Profile

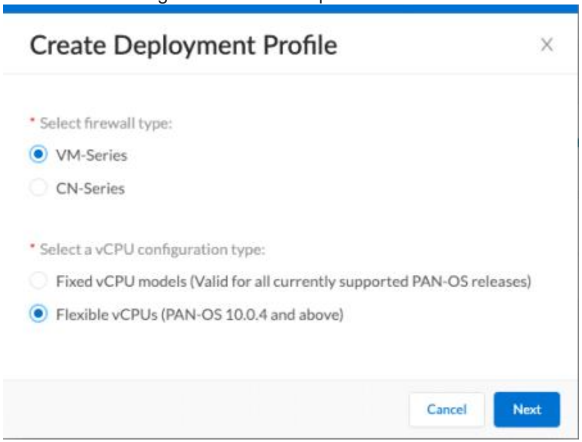
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



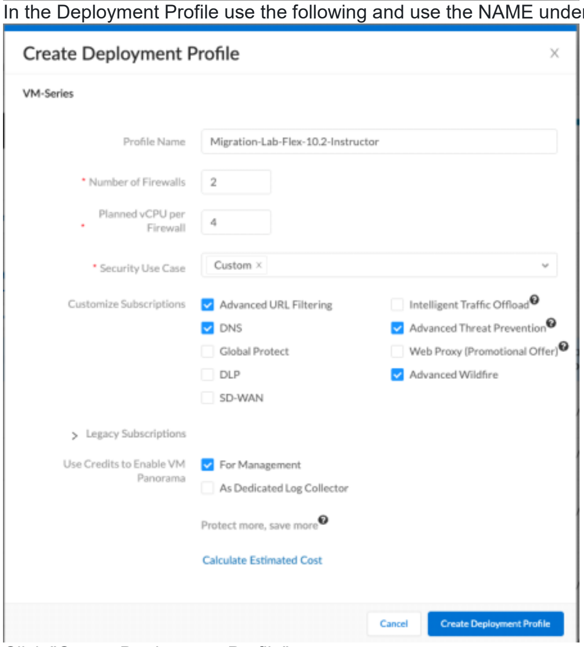
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma NFGW Credits Pool click on Create Deployment Profile



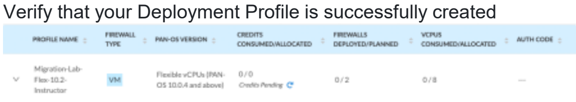
5. Select the following as shown on the picture below and click Next



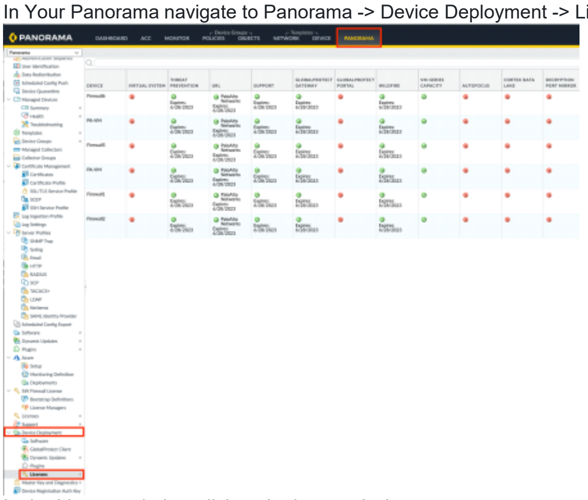
6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-10.2-[StudentName]"



7. Click "Create Deployment Profile"
8. Verify that your Deployment Profile is successfully created

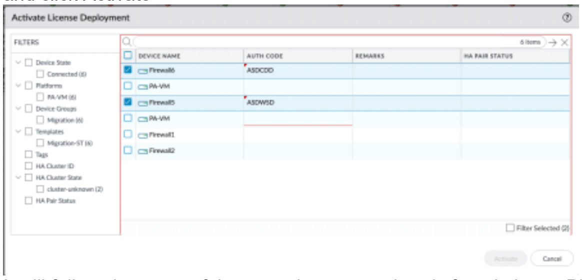


9. Now Copy the Auth Code of the newly created Deployment Profile
10. As next Login in to your Panorama [https://[Public-IP]](https://[Public-IP])
11. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses



12. In the License window click at the bottom Activate

13. Select now Firewall 5 and 6 or as in shown in the Picture. You can verify the Name of the firewalls in the Summary tab. Now type in AUTH CODE field the auth code and click Activate

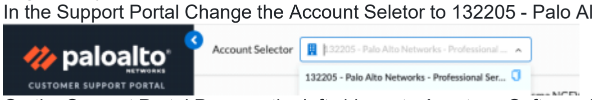


14. It will fail too because of the same issue you already faced above. Please follow the same instructions from the previous chapter to migrate the firewalls to Flexible Deployment profile.

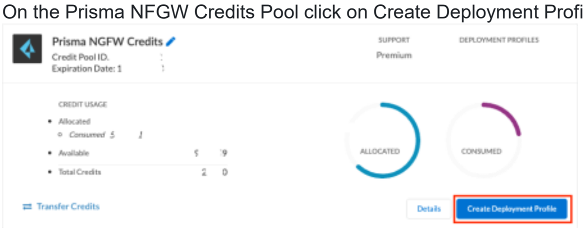
5.3 Change vCPU on PanOS 10.2.3 Firewall

In the following section we will create a new Deployment Profile to change the vCPU on the already licensed Software Firewall

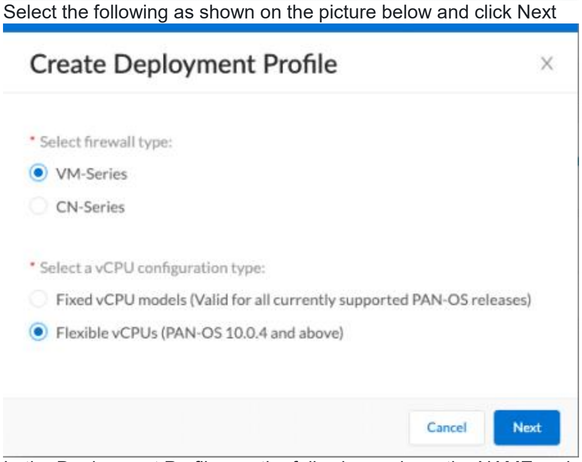
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



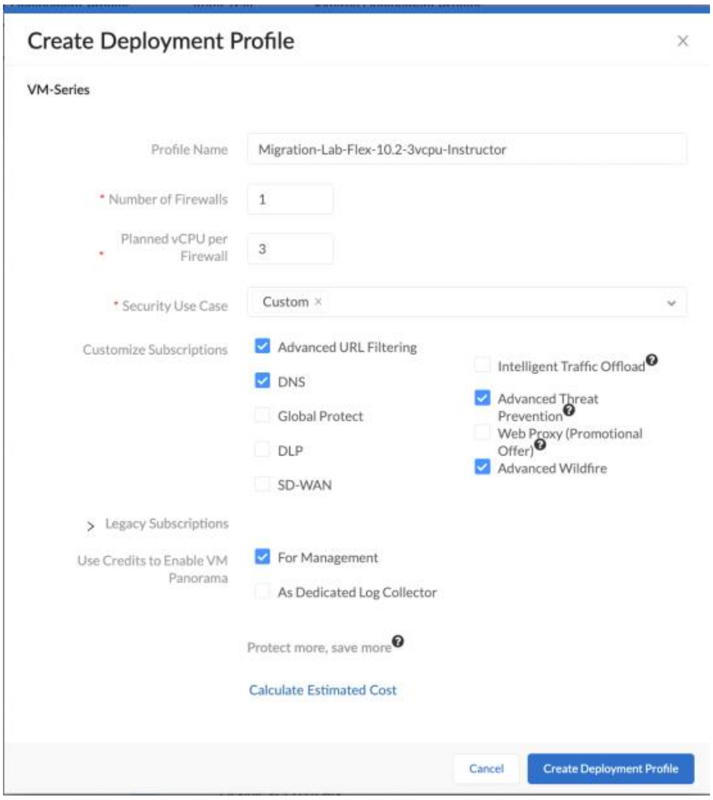
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. On the Prisma NFGW Credits Pool click on Create Deployment Profile



5. Select the following as shown on the picture below and click Next



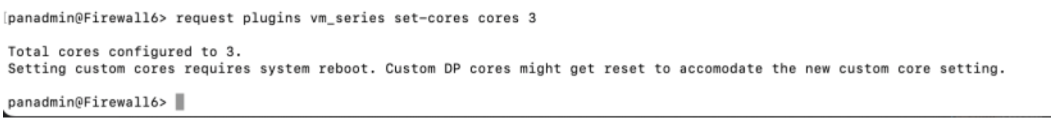
6. In the Deployment Profile use the following and use the NAME under "Profile Name" with "Migration-Lab-Flex-10.2-3vcpu-[StudentName]"



1. Click "Create Deployment Profile"
2. Verify that your Deployment Profile is successfully created

1. Verify at first that both software Firewalls (5 and 6) are migrated to the new Flexible Deployment Profile. Check the Firewall Dashboard if you can see (VM-Series-4)

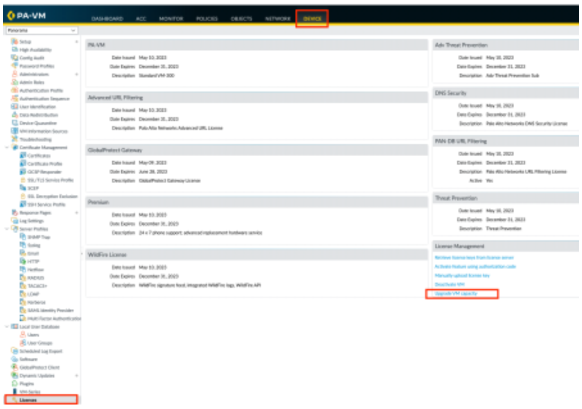
1. As next login to Firewall 5 or 6 via ssh. In my Example i migrate Firewall 6
   ssh -oHostKeyAlgorithms=+ssh-rsa USERNAME@FIREWALL IP
2. In the CLI type the following command to set the Core value to 3
   request plugins vm_series set-cores cores 3



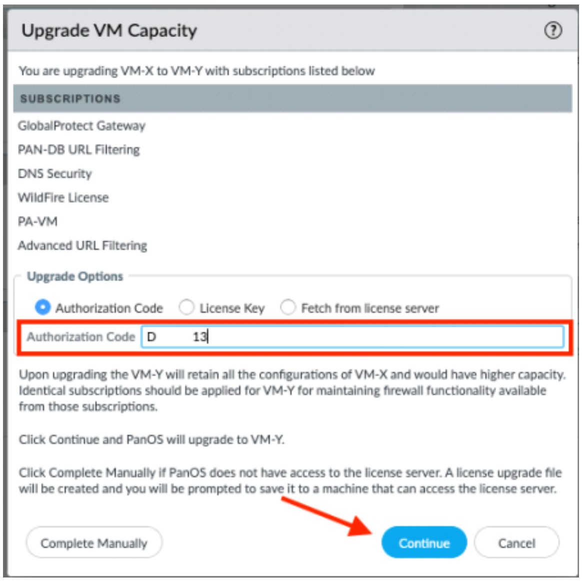
1. The requires a reboot. Type the following command to rebbot the Firewall
   request restart system

```
panadmin@Firewall6> request restart system
Executing this command will disconnect the current session. Do you want to continue? (y or n)

Broadcast message from root (pts/1) (Mon May 15 23:01:05 2023):

The system is going down for reboot NOW!
```

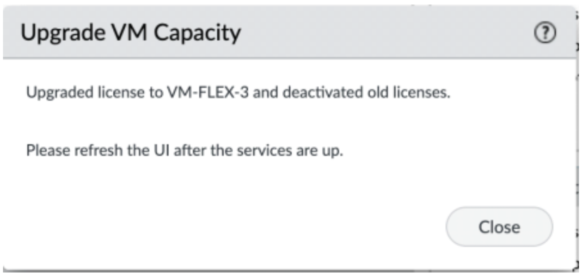
2. The Reboot of the firewall will take now around ~ 5 Minutes
3. Once the Firewall is back online and function login to the Firewall via Panorama or directly to the Firewall
4. In the Firewall navigate Device -> License and click on Upgrade VM capacity



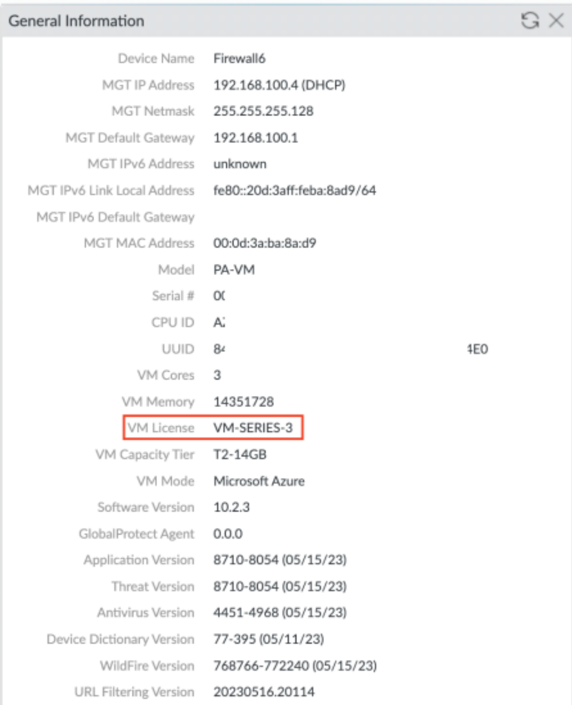
5. In the window add under Authorization Code your auth code (3 vCPU) and click Continue



6. You will see the below outcome once it completed. Click close and refresh the UI



7. In the Firewall go to the Dashboard and you can see the VM License changed to VM-FLEX-3



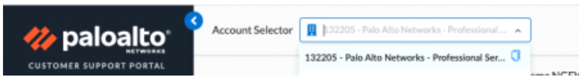
Congratulations!!! You successful migrated 1 Software Firewalls from Flexible Deployment Profile with 4 vCPU's to a Flexible Deployment Profile and changed the Cores count via CLI

5.4 Change/Update Deployment Profiles

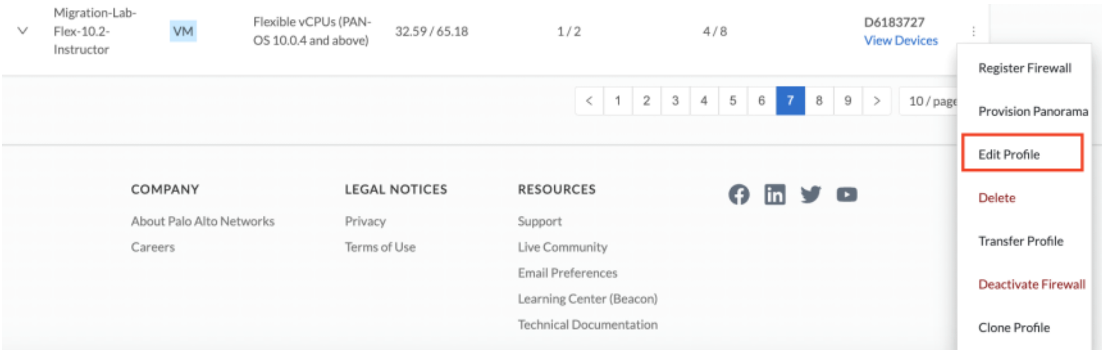
In the following section you will now update your Deployment Profile ("Migration-Lab-Flex-10.2-[StudentName]") too remove some Subscription and enable subscriptions

5.4.1 Add Security Subscriptions

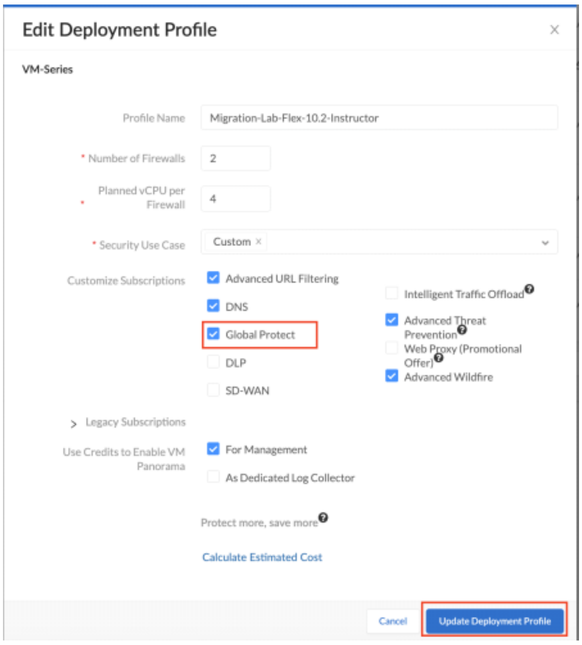
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



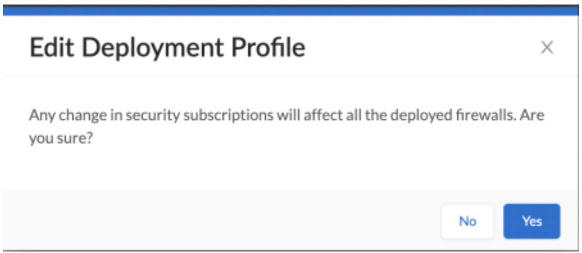
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. Go to youe Deployment Profile "Migration-Lab-Flex-10.2-[StudentName]" click in the three dots and Edit Profile



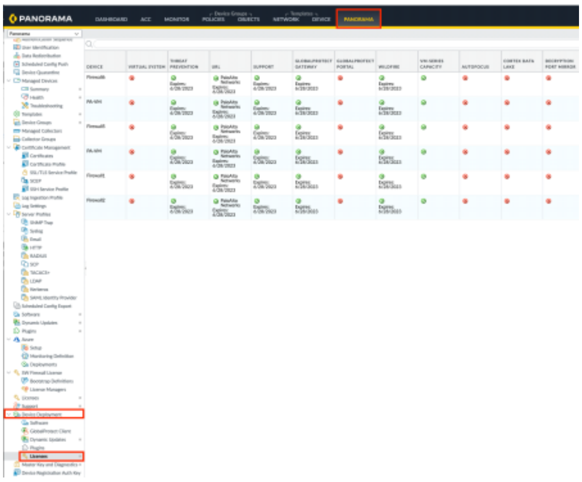
5. In your Deployment Profile select the Global Protect and Click Update Deployment Profile



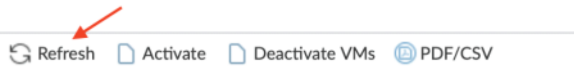
6. Click YES in the new Window



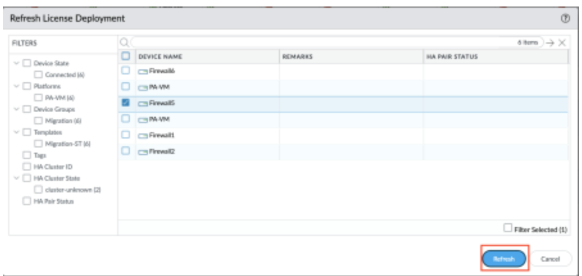
7. As next Login to your Panorama
8. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses



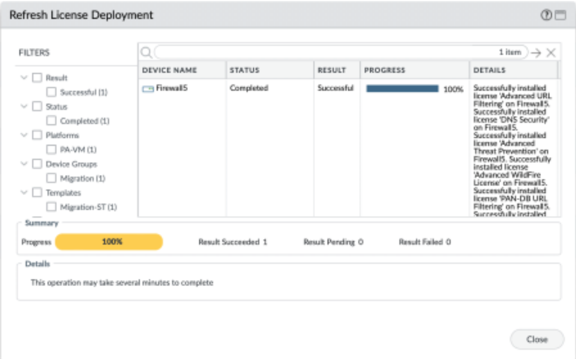
9. Select Refresh



10. Now Select the firewall who was associated with the Auth Code of the "Migration-Lab-Flex-10.2-[StudentName]" Deployment Profile (In the Example is it Firewall 5) and click Refresh



11. You should see the following output if it was successfull



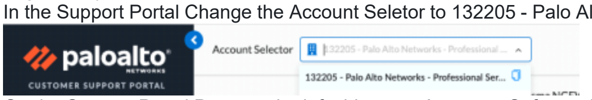
12. Refresh the Panorama UI

13. Now you should see that on Software Firewall 5 is the Global Protect License Active

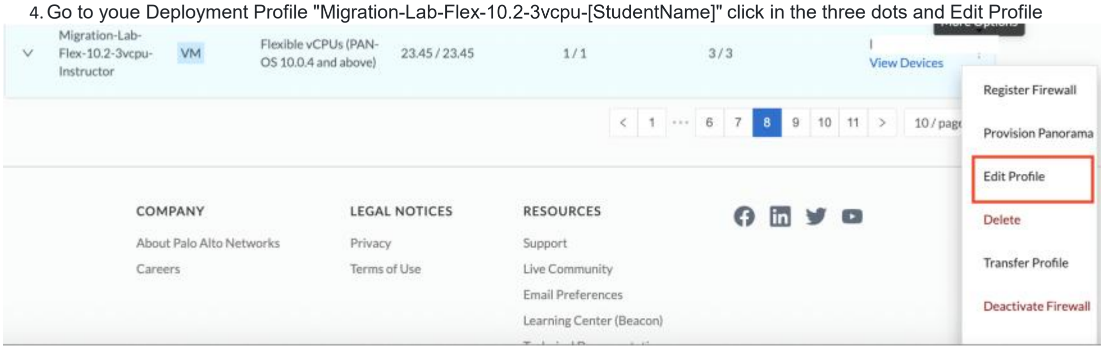
Congratulations!!! You successful Updated your Deployment Profile and added another Security subscription

5.4.2 Remove Security Subscriptions

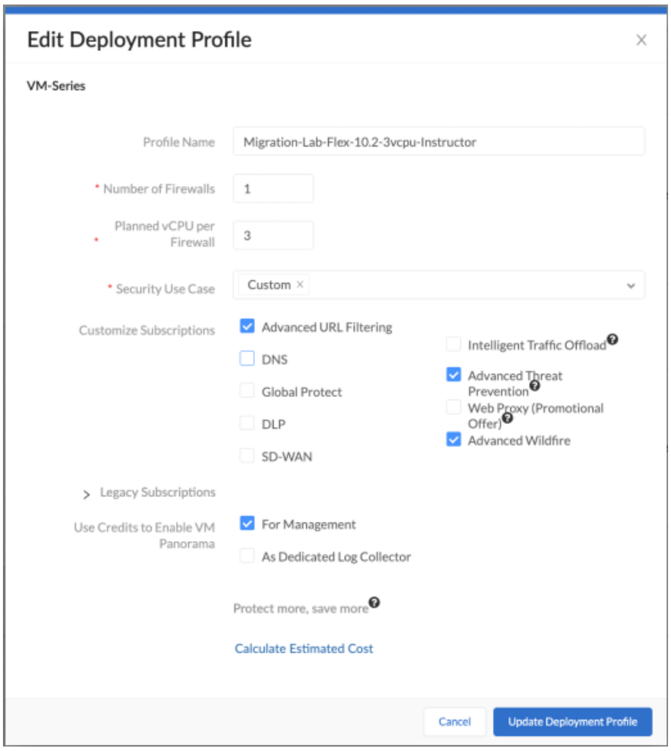
1. Login with your PANW Credentials at the Customer Support Portal https://support.paloaltonetworks.com/
2. In the Support Portal Change the Account Seletor to 132205 - Palo Alto Networks - Professional Services



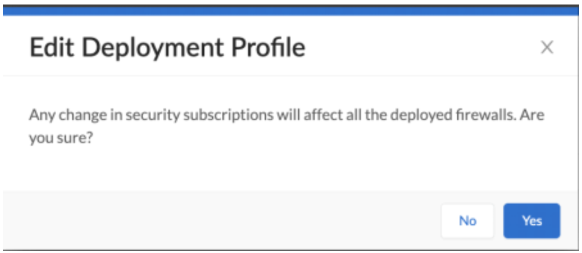
3. On the Support Portal Page on the left side go to Assets -> Software NGFW Credits
4. Go to youe Deployment Profile "Migration-Lab-Flex-10.2-3vcpu-[StudentName]" click in the three dots and Edit Profile



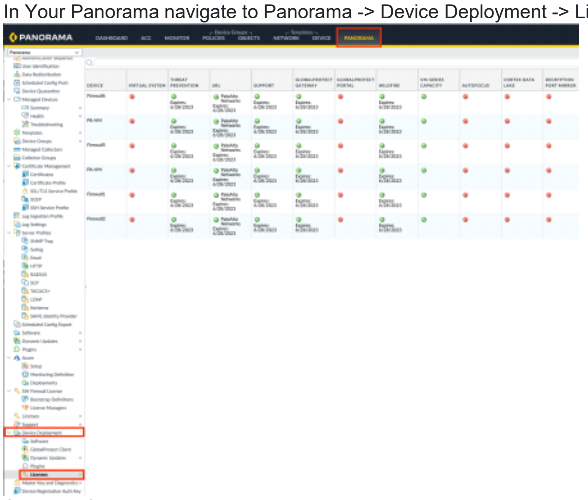
1. In your Deployment Profile de-select the DNSt and Click Update Deployment Profile



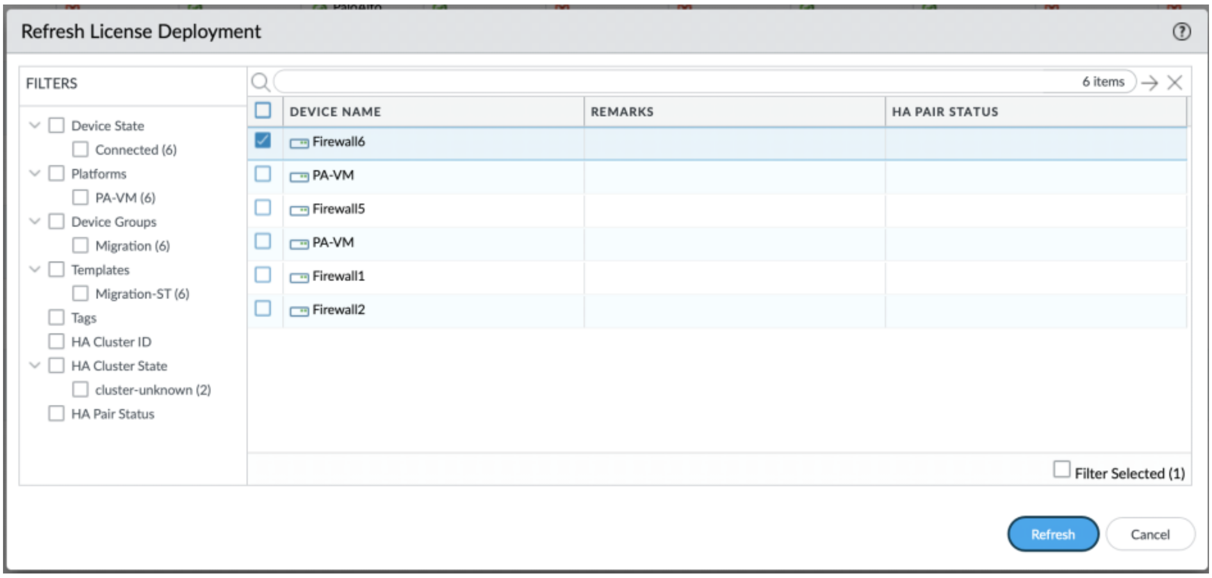
1. Click YES in the new Window



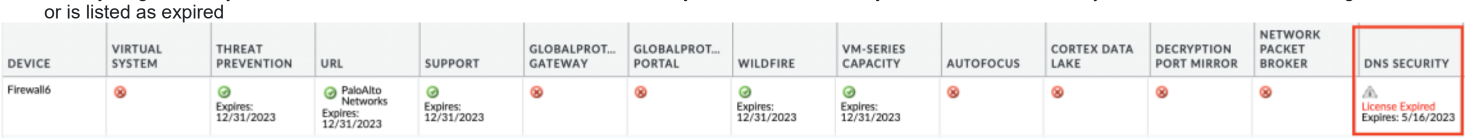
2. As next Login to your Panorama
3. In Your Panorama navigate to Panorama -> Device Deployment -> Licenses



4. Select Refresh
5. Now Select the Firewall who is associated with the Update Auth Code.



1. Are the Update working?
2. If everything Worked you have to refresh the UI. In some cases it will show you an error but when you check the Panorama you can see the DNS license got removed or is listed as expired



DEVICE	VIRTUAL SYSTEM	THREAT PREVENTION	URL	SUPPORT	GLOBALPROT... GATEWAY	GLOBALPROT... PORTAL	WILDFIRE	VM-SERIES CAPACITY	AUTOFOCUS	CORTEX DATA LAKE	DECRYPTION PORT MIRROR	NETWORK PACKET BROKER	DNS SECURITY
Firewall6		Expires: 12/31/2023	PaloAlto Networks Expires: 12/31/2023	Expires: 12/31/2023			Expires: 12/31/2023	Expires: 12/31/2023					License Expired Expires: 5/16/2023

Congratulations!!! You successful Updated your Deployment Profile and removed a Security subscription

From <https://github.com/PaloAltoNetworks/flex-license-migration-lab>

Palo-Alto VM using VMWare Workstation

Thursday, December 25, 2025 8:40 PM

OVA Download link: https://upw.io/4A9/PA-VM-ESX-10.1.0.ova

Hypervisor
- VMWare workstation (latest version)
- Oracle Virtual Box (latest version)

Default username: **admin**
Default password: **admin** (note - you might have to login 4-5 times with the default password)

To view the system details
show system info

Other commands:

Basic System Commands
show system info show system state show system environmentals show system resources show system disk-space show system software status show system uptime show system files

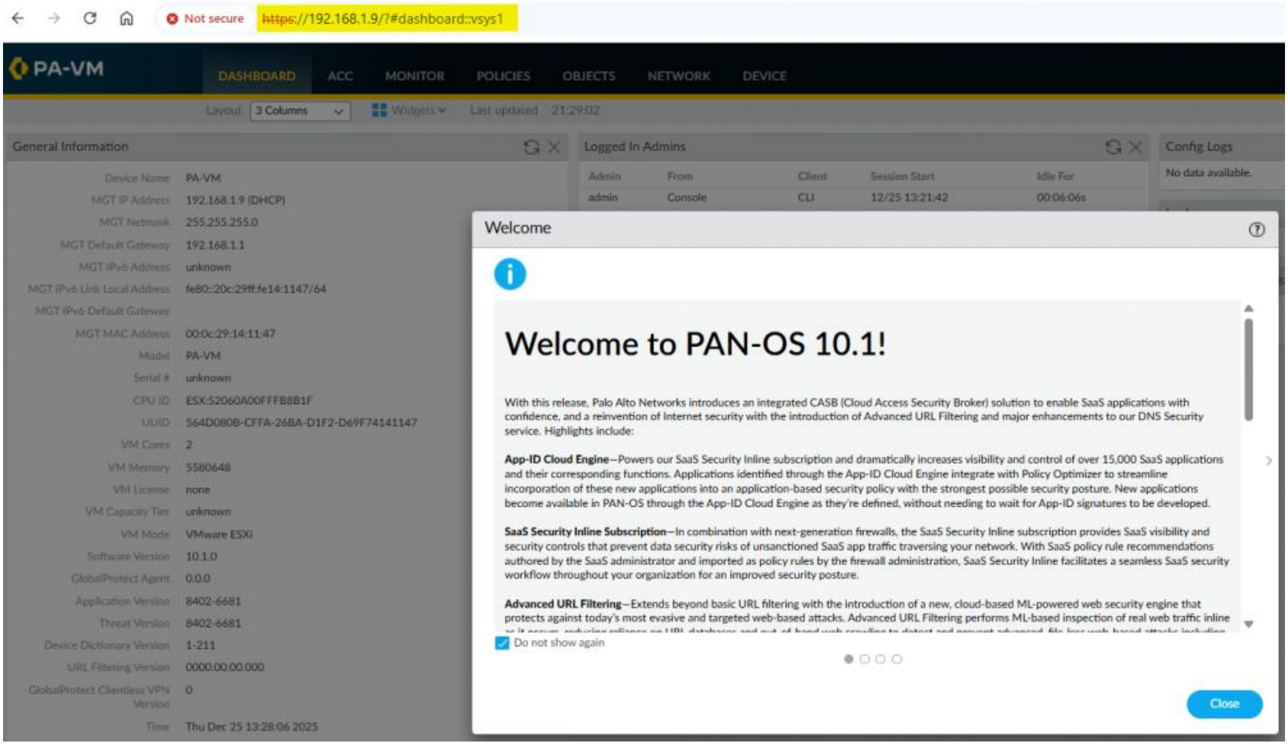
Interface & Networking Commands
show interface all show interface ethernet1/1 show interface management show routing route show routing protocol bgp summary show routing protocol ospf neighbor show arp all show arp interface ethernet1/1 ping host <ip-address> traceroute host <ip-address>

Policy & Security Rule Commands
show running security-policy show running nat-policy show running qos-policy show running decryption-policy show running application-override-policy

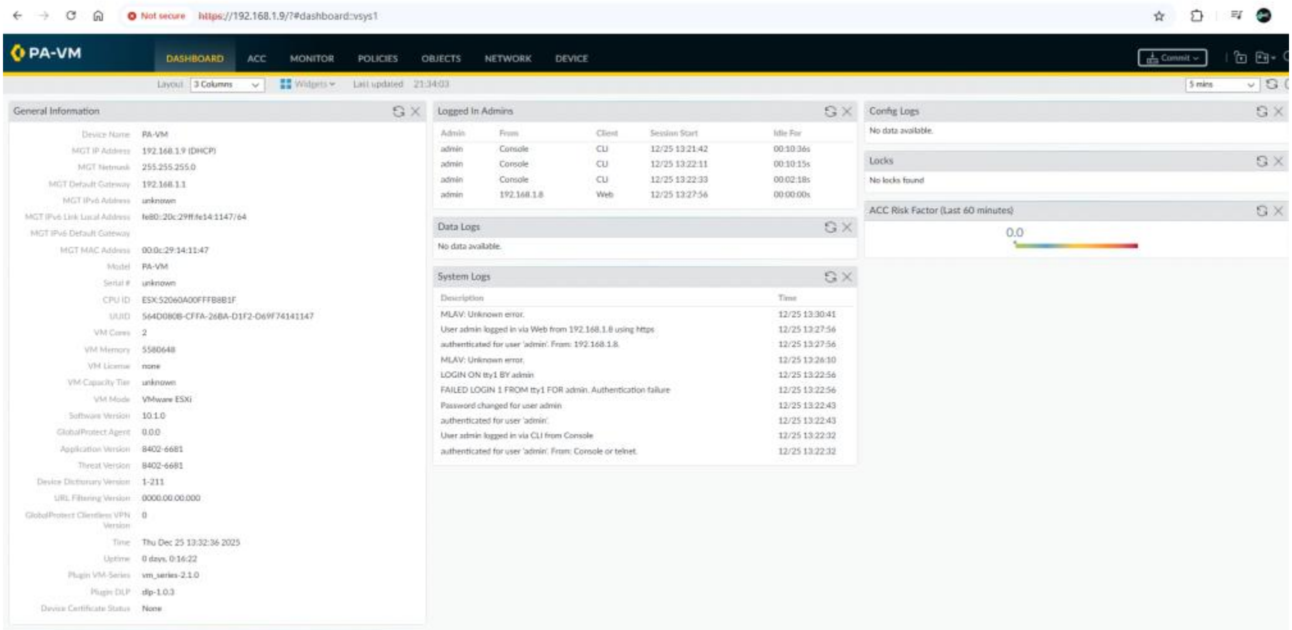
Traffic, Session & Application Commands

```
show session all
show session id <session-id>
show session info
show session statistics
show counter global
show counter interface
show application all
show application statistics
```

Accessing PA using web browser:



Dashboard:



Palo-Alto VM using Oracle VirtualBox

Sunday, December 28, 2025 8:57 AM

1. Prerequisites and Planning

1.1 Supported Palo Alto VM Image

Use VM-Series for VMware (OVA)

(Palo Alto does not officially provide a VirtualBox image; VirtualBox works but is best-effort only).

1.2 Host System Requirements

- CPU with hardware virtualization enabled (Intel VT-x / AMD-V)
- Minimum 8 GB RAM (16 GB recommended)
- At least 60 GB free disk space
- Oracle VirtualBox 7.x (latest recommended)
- VirtualBox Extension Pack installed

1.3 Files Required

Palo Alto VM-Series .ova file

Internet access (optional but recommended for updates)

2. Importing the Palo Alto OVA into VirtualBox

Step 1: Launch VirtualBox

- Open **Oracle VirtualBox Manager**

Step 2: Import Appliance

a. Click **File → Import Appliance**
b. Browse and select the **Palo Alto .ova file**
c. Click **Next**

Step 3: Review and Modify Appliance Settings

Carefully validate and adjust the following:

3.1 System → Motherboard

- **Base Memory**: 8192 MB (minimum)
- **Boot Order**: Hard Disk first
- **Chipset**: PIIX3
- **EFI**: ✘ Disabled

3.2 System → Processor

- **CPUs**: 4 (minimum)
- **Enable PAE/NX**: ☑ Enabled

3.3 Display

- **Graphics Controller**: VMSVGA
- **Video Memory**: 128 MB

Avoid VBoxVGA or VBoxSVGA — they often cause boot instability.

3.4 Storage

- Confirm the **virtual disk is attached** (SATA controller)
- Do **not** change disk format

3.5 Network (Critical)

Configure **at least two adapters**:

Adapter	**Purpose**	**Setting**
Adapter 1	Management (mgmt)	Bridged Adapter **or** Host-Only

Adapter 2 Data Plane Internal Network

- Adapter Type: **Intel PRO/1000 MT Desktop**
- Promiscuous Mode: Allow All

Click **Finish** to import.

Login Credentials

- Username: admin
- Password: admin

Fix the kernel panic Issue (Step-by-Step – Proven)

Step 1: Power Off the VM

Ensure the VM is **completely powered off**.

Step 2: Remove Incorrect Disk Controllers

1. Open **VM Settings → Storage**
2. Remove:
   - IDE Controller
   - SCSI Controller (if present)
3. **Do NOT delete the disk file**

Step 3: Add Correct Controller (AHCI SATA)

1. Click **Add Controller**
2. Select **SATA Controller**
3. Set:
   - Name: SATA Controller
   - Type: **AHCI**
   - Ports: 2

Step 4: Attach Palo Alto Disk to SATA

1. Under SATA Controller → **Add Hard Disk**
2. Choose **Existing Disk**
3. Select the imported .vmdk
4. Ensure:
   - **Port 0**
   - **Device 0**

Step 5: Confirm Boot Order

System → Motherboard

- Boot Order:
  1. Hard Disk
  2. Optical
- Disable everything else

Step 6: Start the VM

Expected boot sequence:

- GRUB loads
- Kernel loads
- Root filesystem mounts
- Login prompt appears

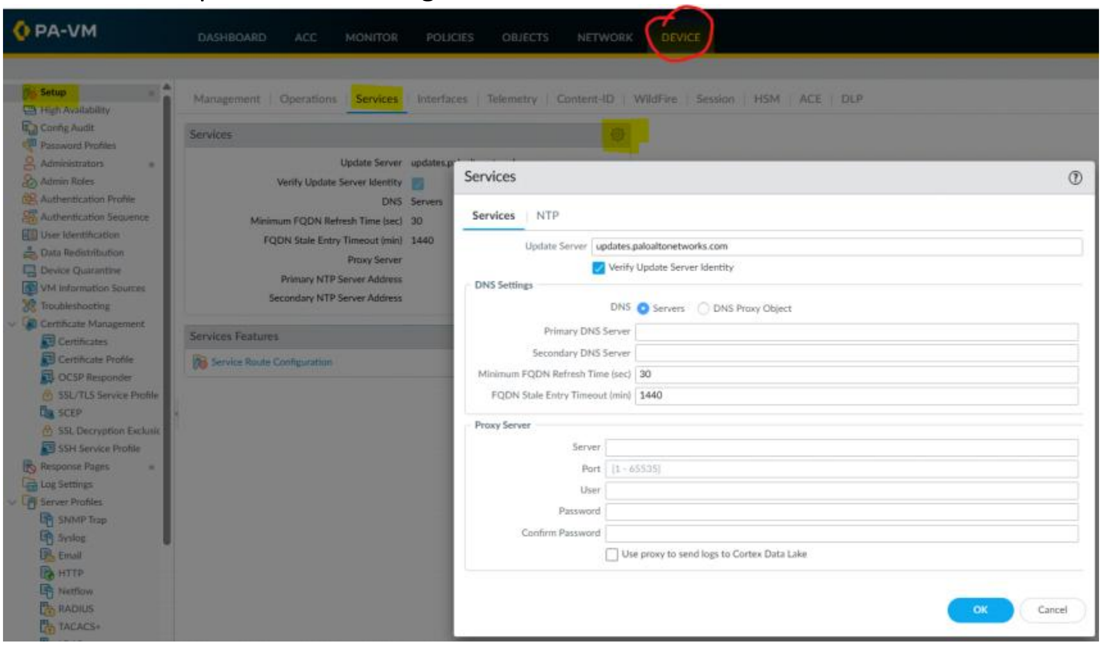
Note - Palo Alto VM-Series on VirtualBox will only boot reliably if the root disk is attached to an AHCI SATA controller, Port 0, using the VMware OVA image.

Configuring DNS & NTP

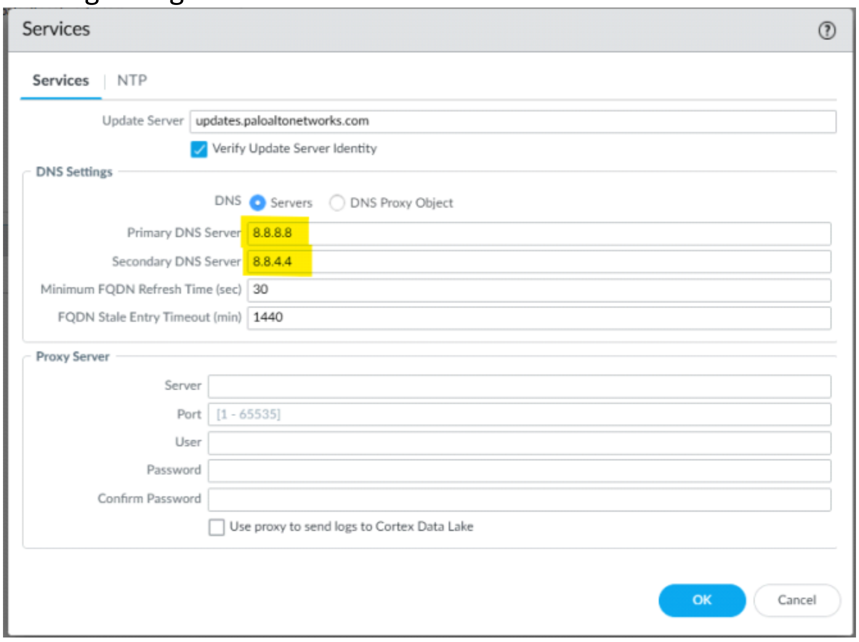
Thursday, December 25, 2025 9:09 PM

Domain Name System (DNS)

- It helps in translating/resolving name (ex: www.example.com) into valid IP address (ex: 1.1.1.1)
- Device --> Setup --> Services --> gear icon:

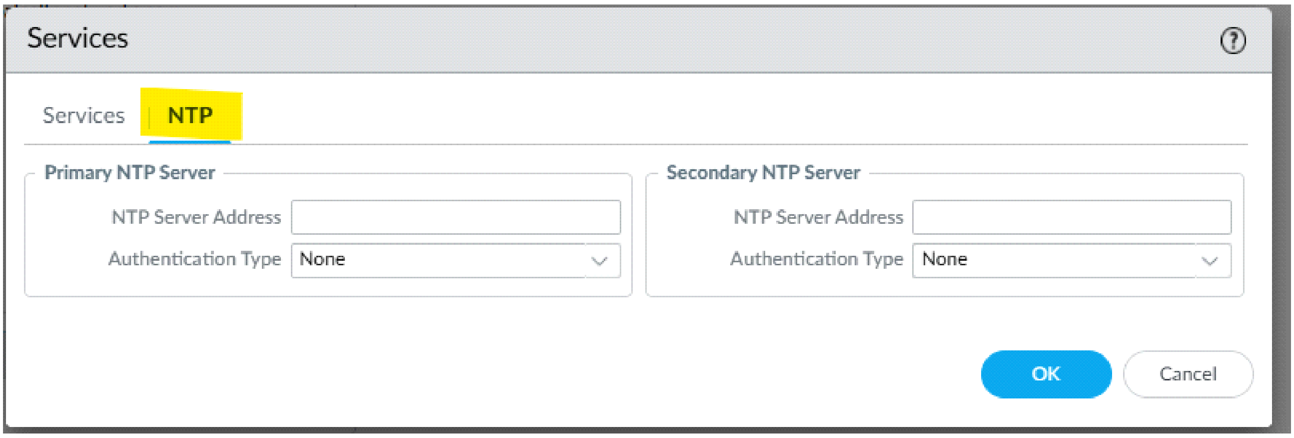


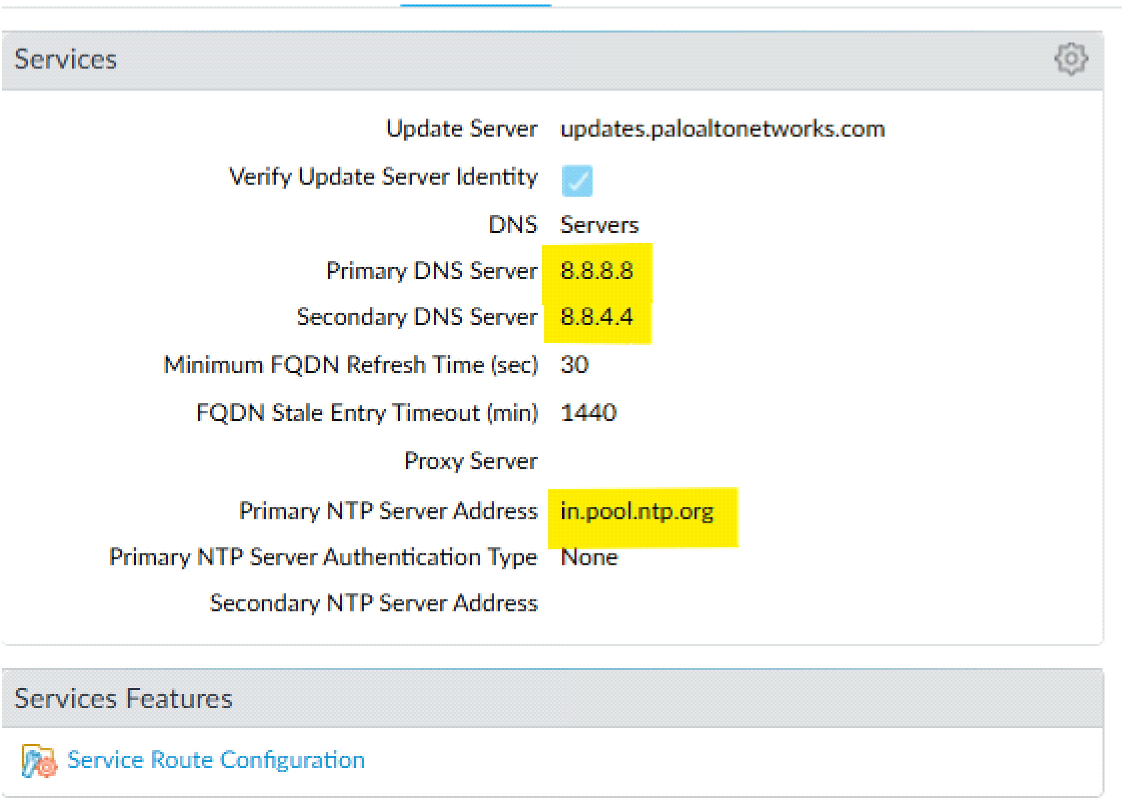
Making changes:



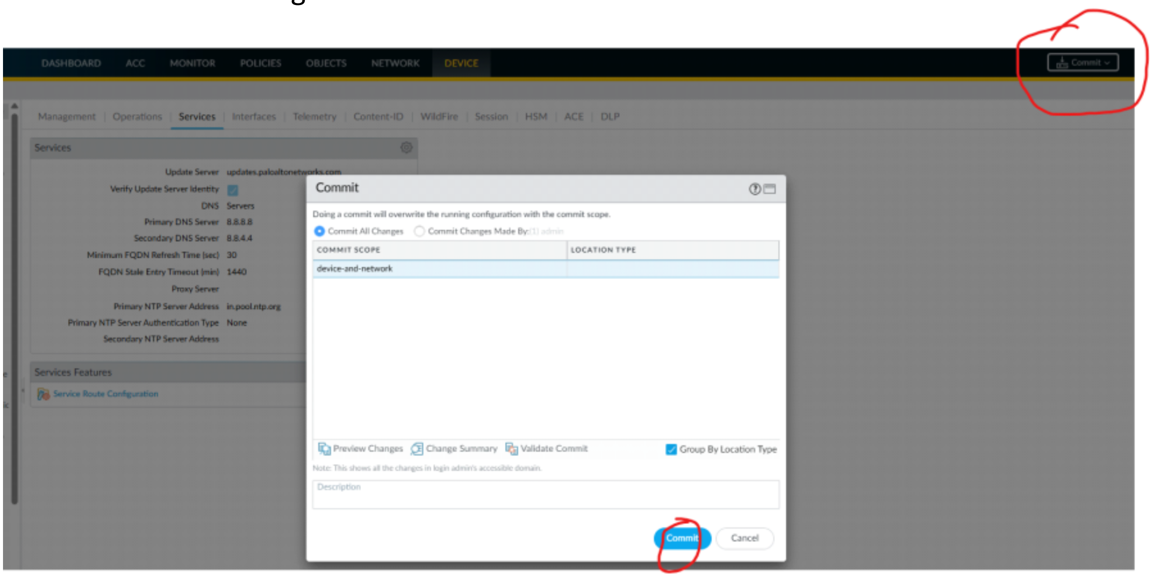
Network Time Protocol (NTP)

- It's a networking protocol used for synchronizing clock between computers over the network.
- Changing NTP config:





Now commit the changes:

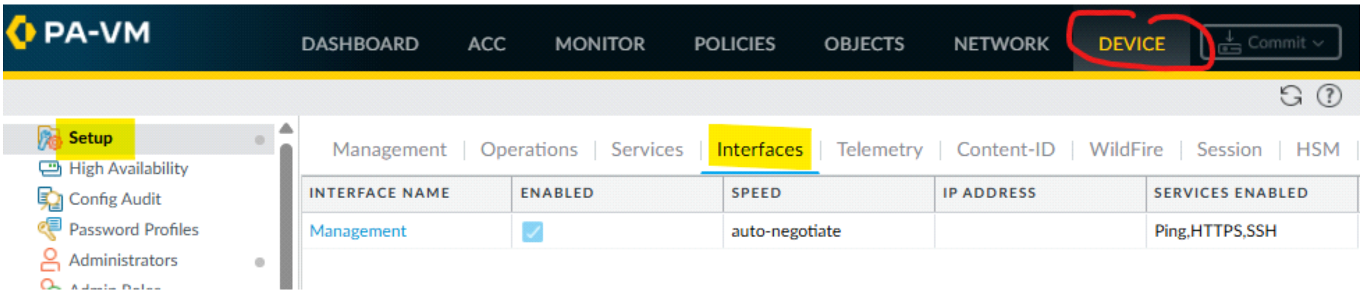


Ping google using management network:
ping host google.com

Managing interfaces using PA-FW

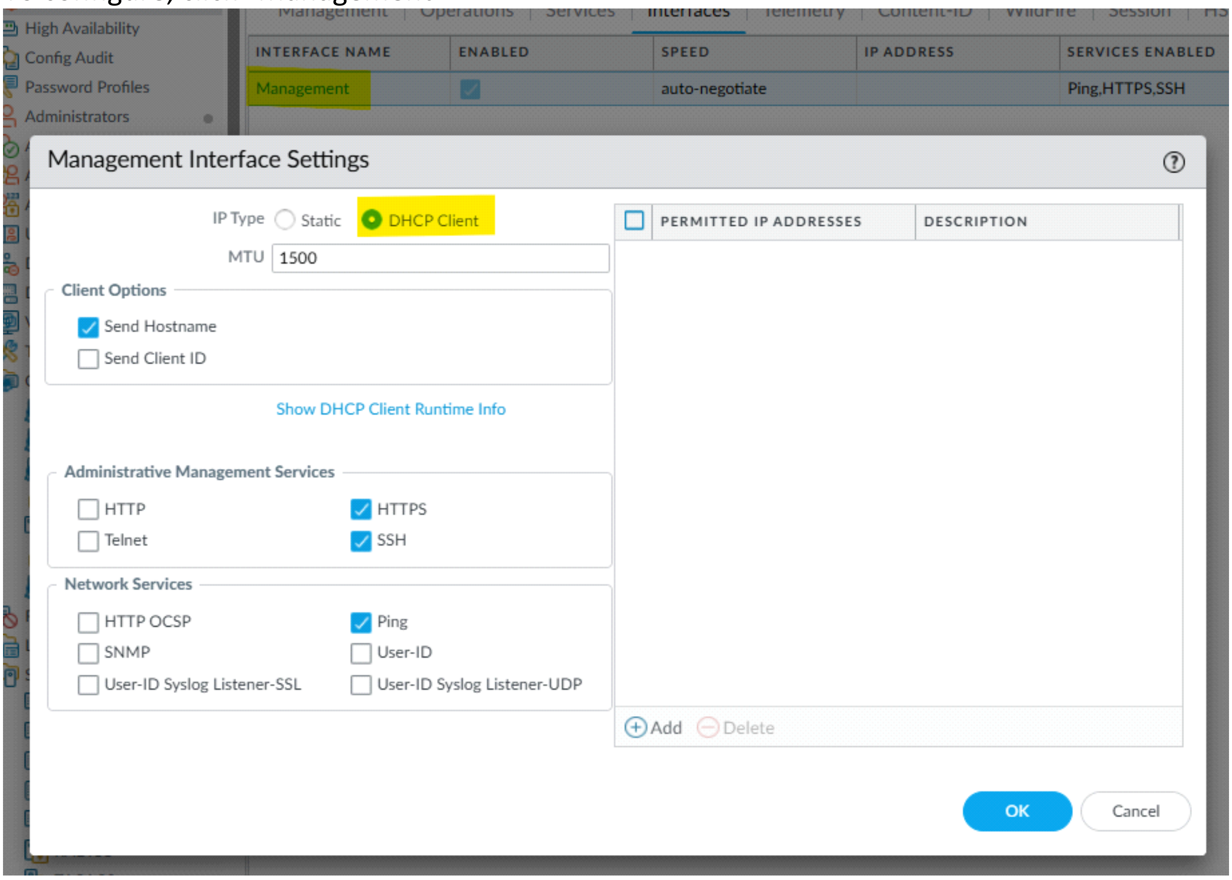
Thursday, December 25, 2025 10:30 PM

Go to interfaces:



- It allows ping, ssh and HTTPS connections. Rest all will be denied (as of now, but can be changed).

To configure, click "Management"



You can make changes here and then commit it.

To do the same using CLI:

```
admin@PA-VM> show system services

HTTP          : Disabled
HTTPS         : Enabled
Telnet        : Disabled
SSH           : Enabled
Ping          : Enabled
SNMP          : Disabled
```

Switch to config mode by typing "configure" :

To remove the dhcp configuration:

- # delete deviceconfig system type dhcp-client

To add static configuration:

- # set deviceconfig system type static

To provide static IP address with subnet mask

- # set deviceconfig system ip-address 192.168.1.100 netmask 255.255.255.0

To finalize the configuration, we need to commit the changes.

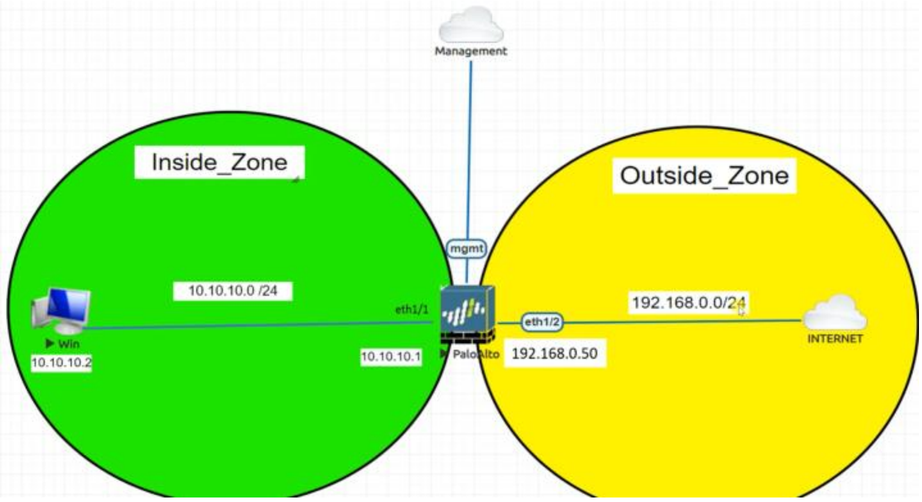
- # commit

Basic requirements - firewall connectivity

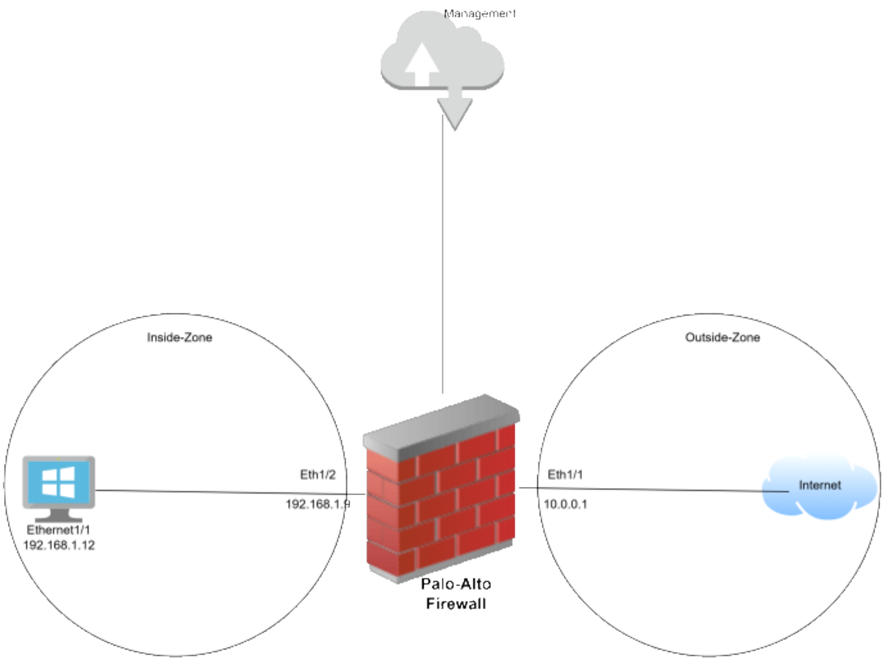
Thursday, December 25, 2025 10:59 PM

Below are the basic requirements to establish the connectivity through firewall:

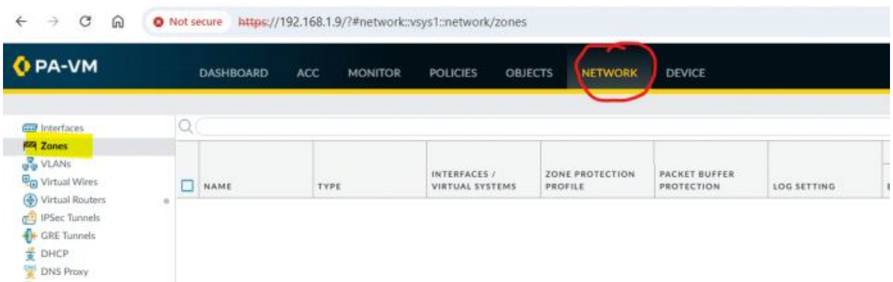
1. Define the firewall zones.
2. Configure the firewall interfaces
3. Configure routing to route the traffic between zones.
4. Define the NAT policy.
5. Define the security policy either to permit or deny.



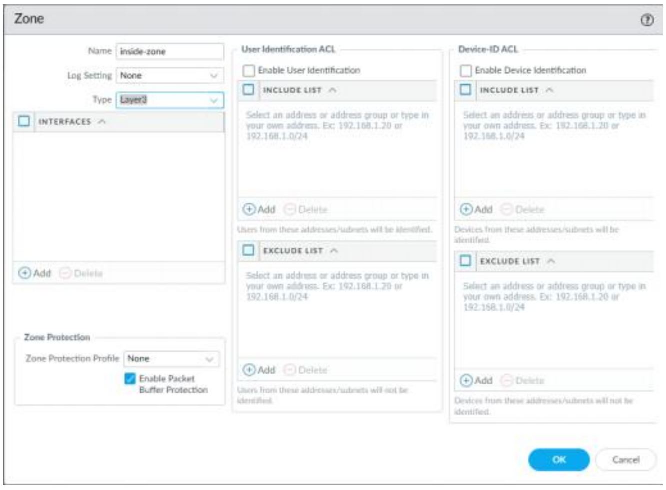
My lab setup:



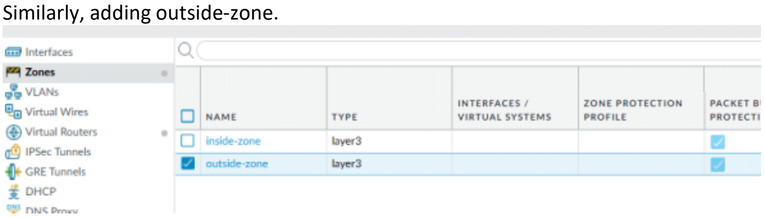
Create zones in PA-FW:



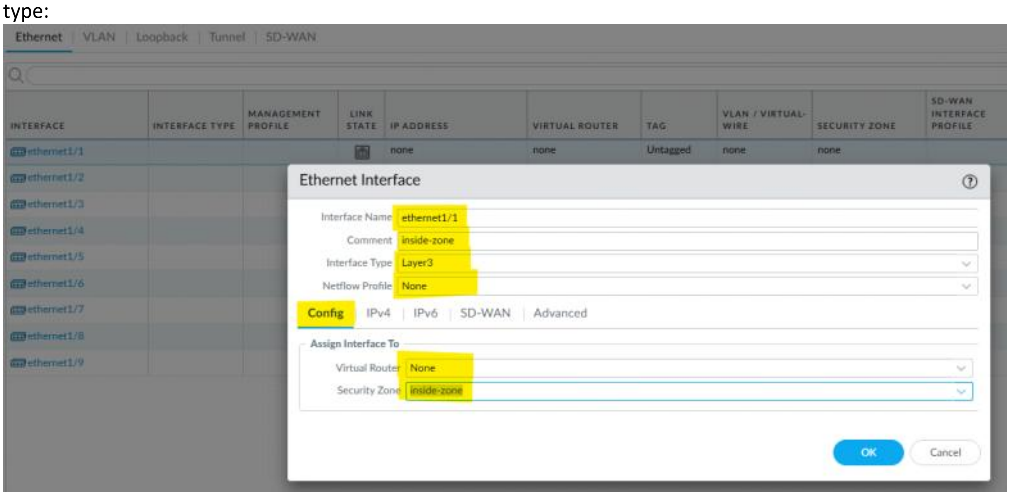
Then click on "Add" at the bottom to add zone(s).



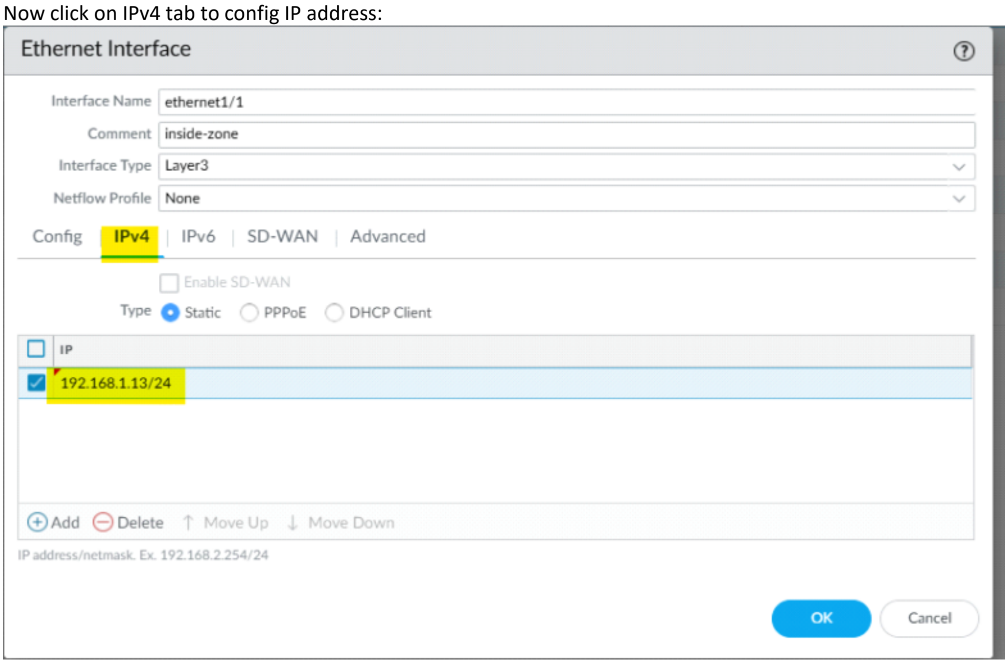
Similarly, adding outside-zone.



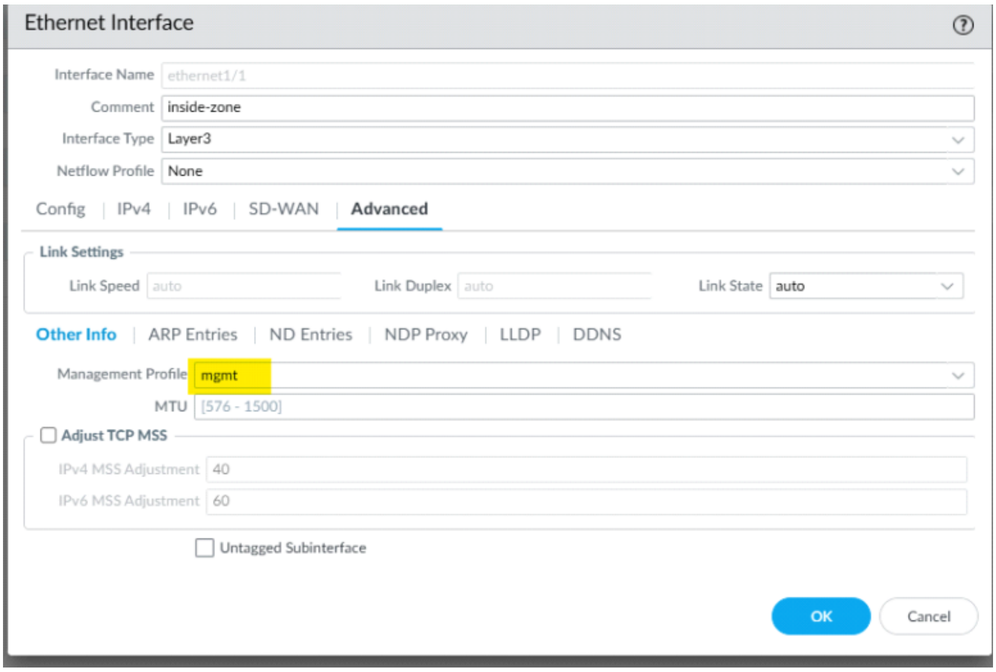
Go to Network -> Interfaces -> Ethernet -> Click on "ethernet1/1" and fill comment and int type:



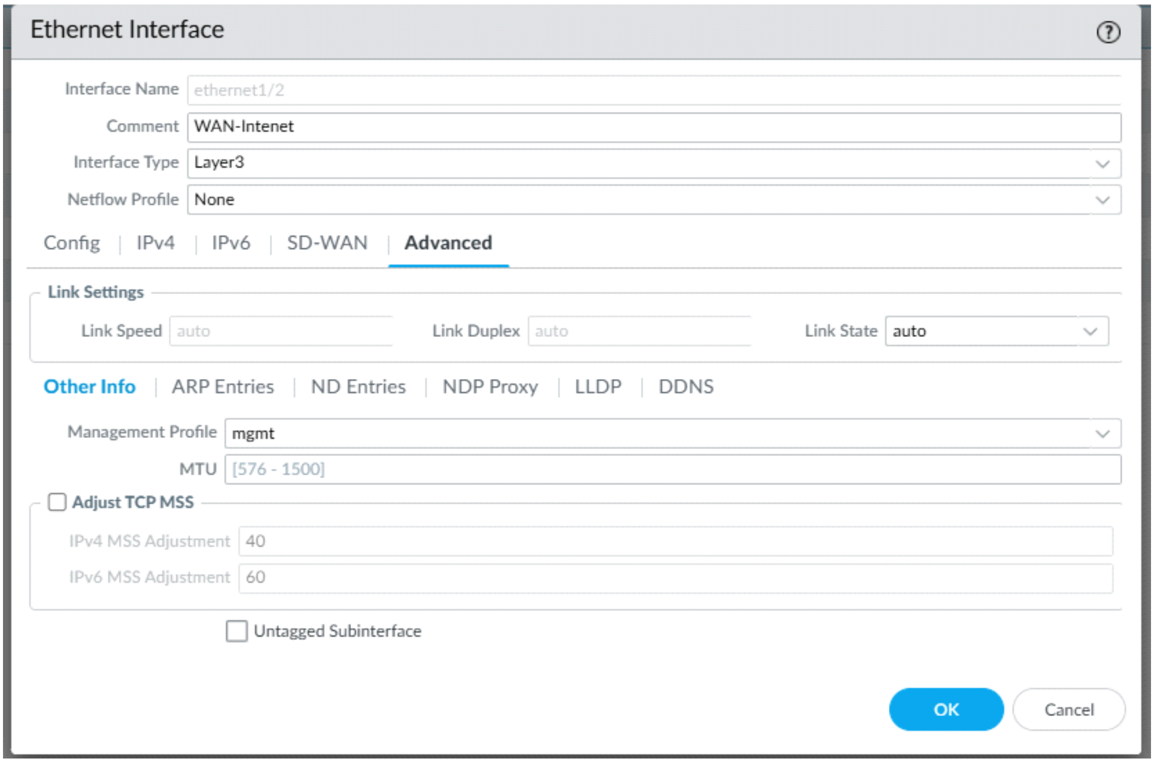
Now click on IPv4 tab to config IP address:



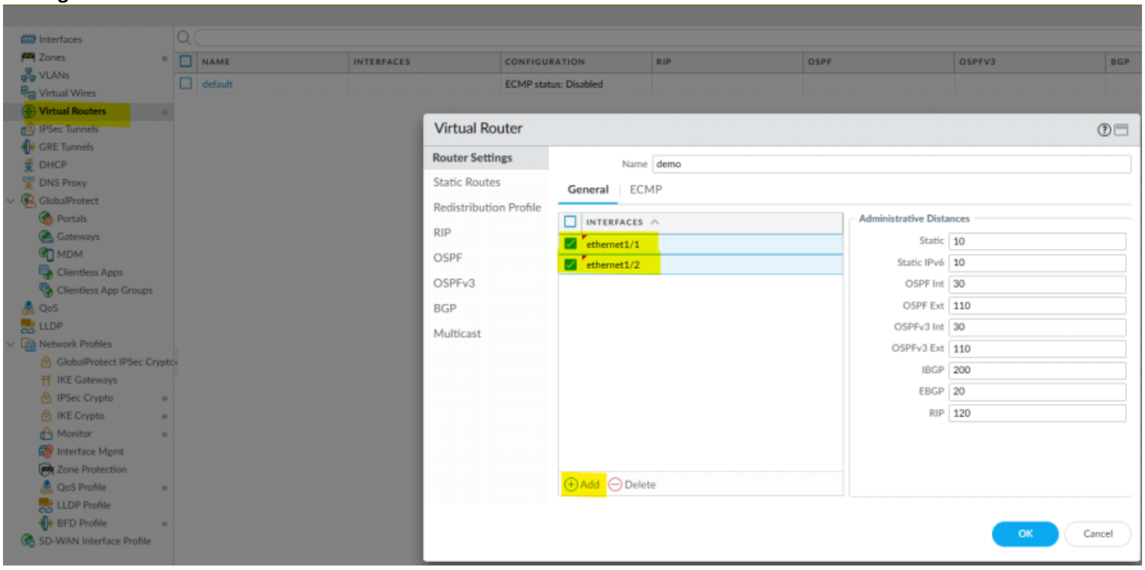
Now to go to "Advanced" and create a management port for ping (not recommended for production):



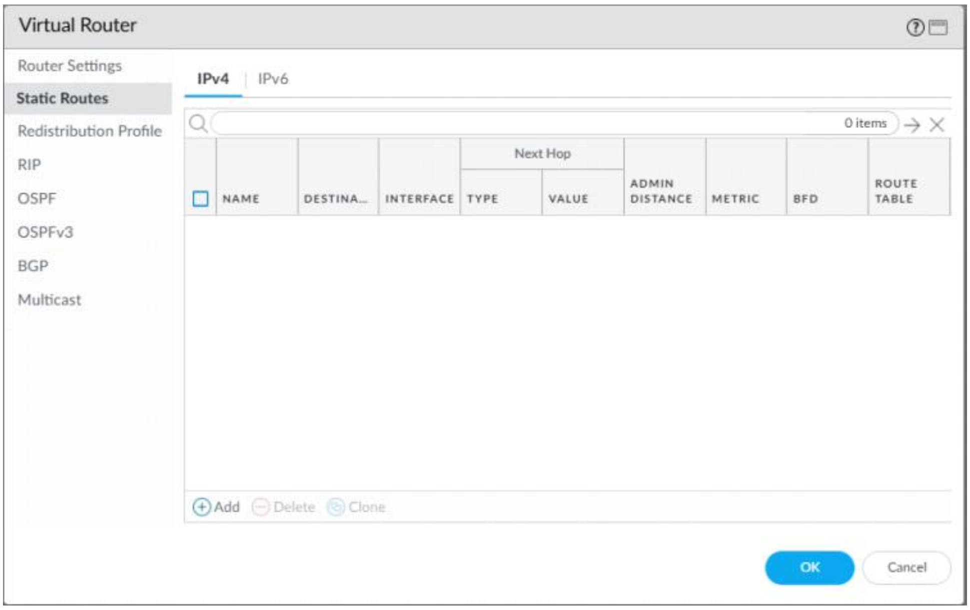
Now configuring the "Ethernet 1/2" for internet:



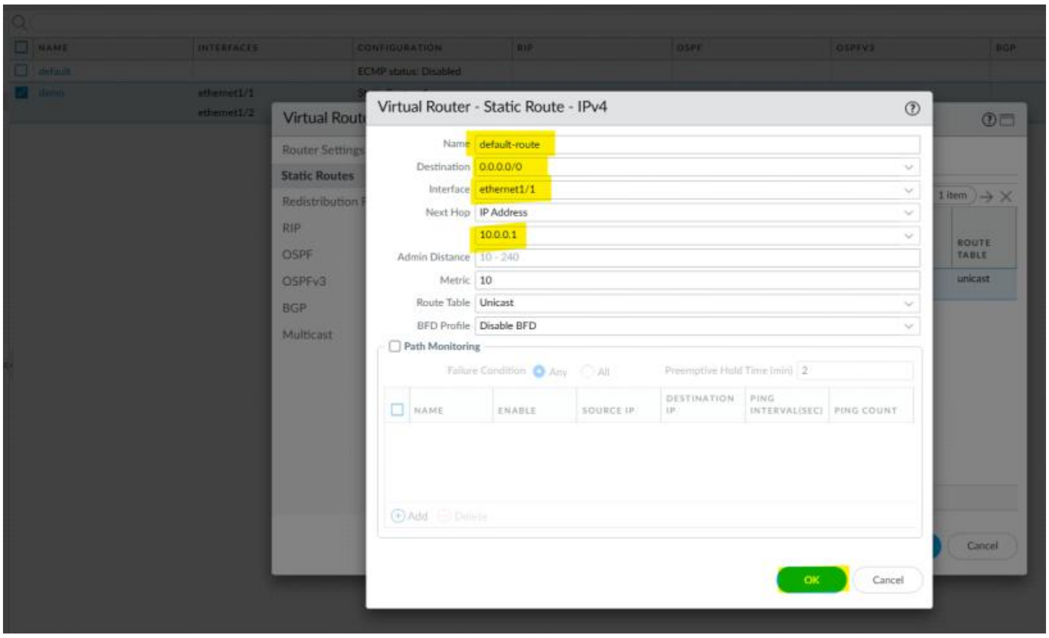
Adding "Virtual Router" with both interfaces:



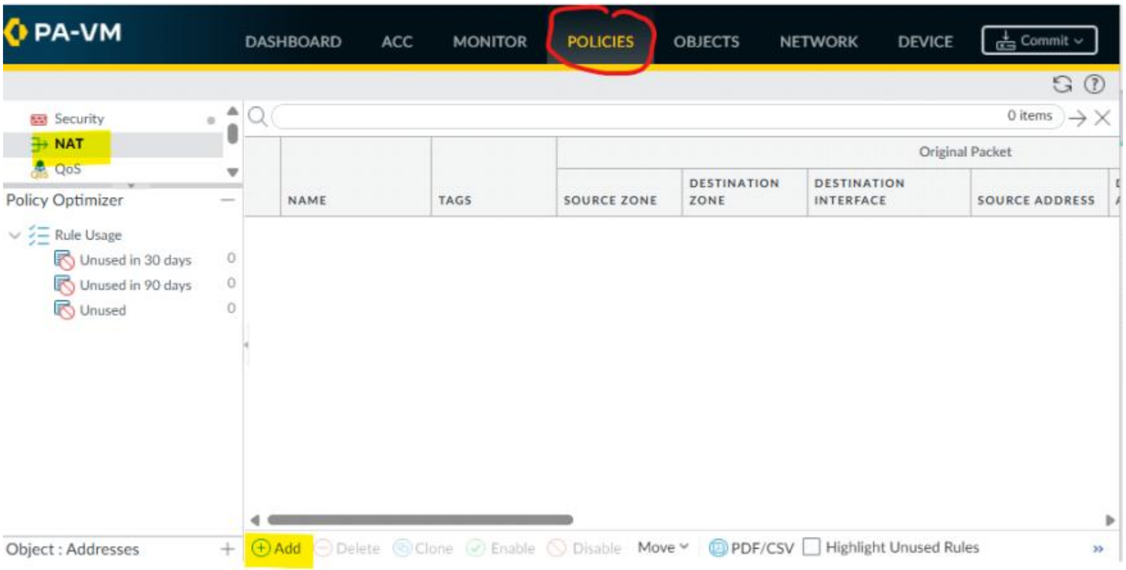
Select "static routes"



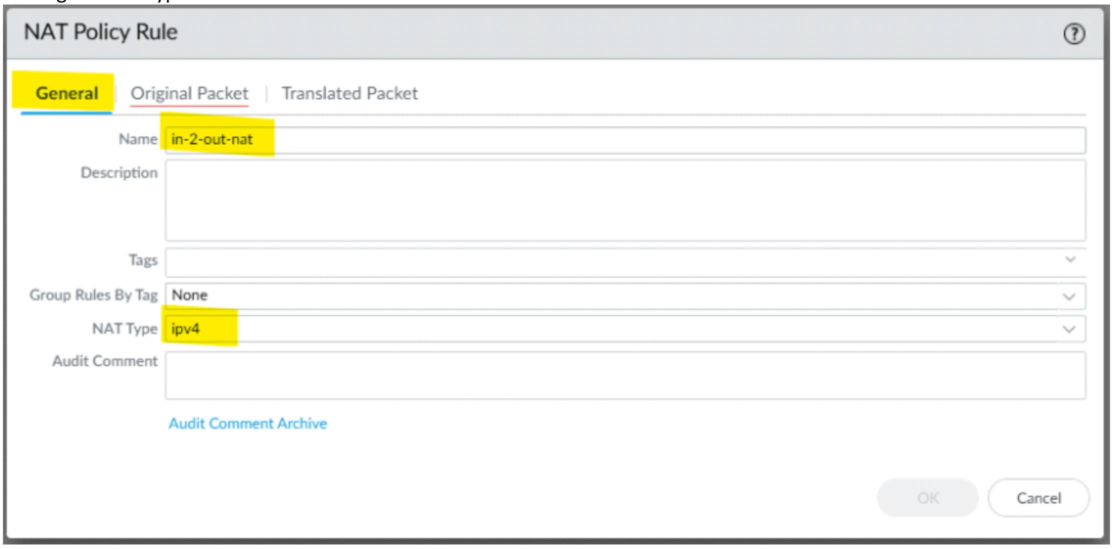
Click "add"



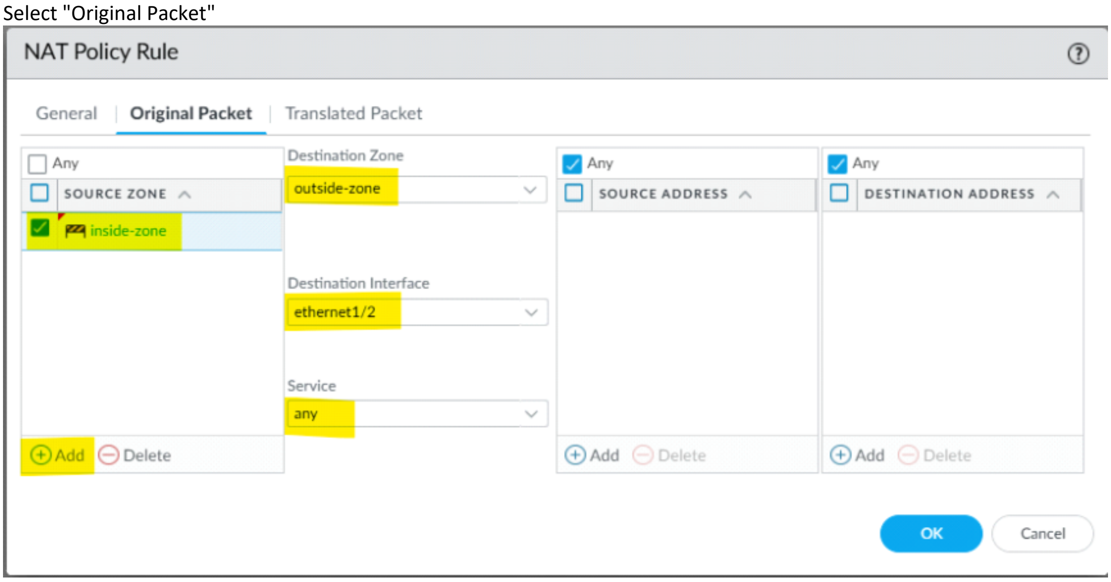
Now configuring the NAT Policy:



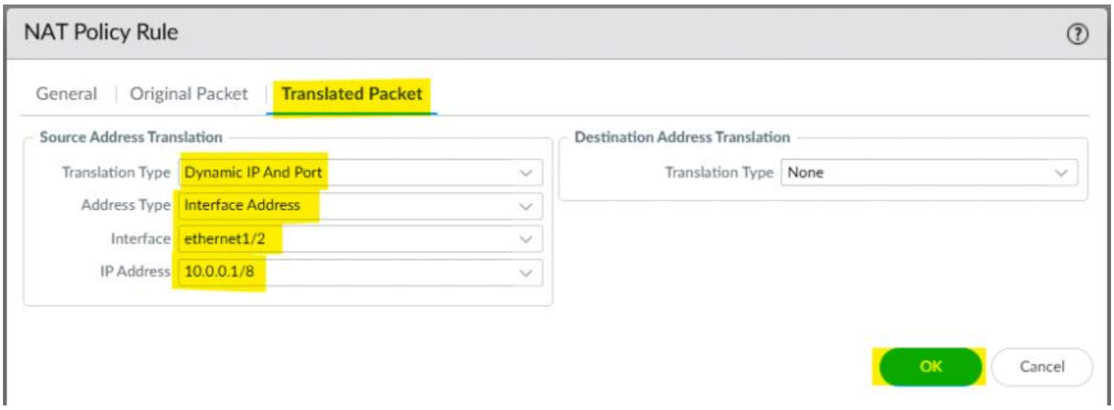
Adding name & type:

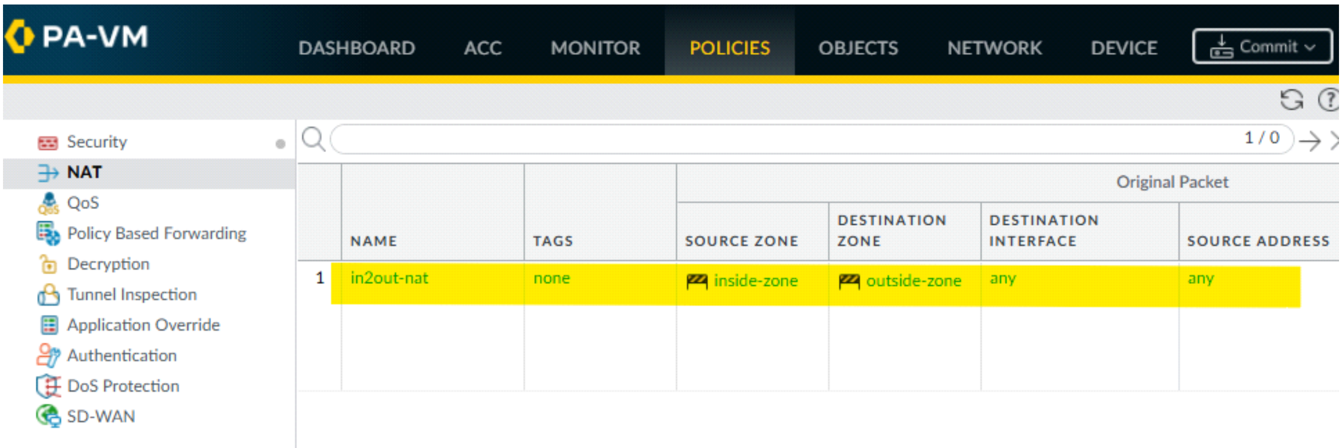


Select "Original Packet"

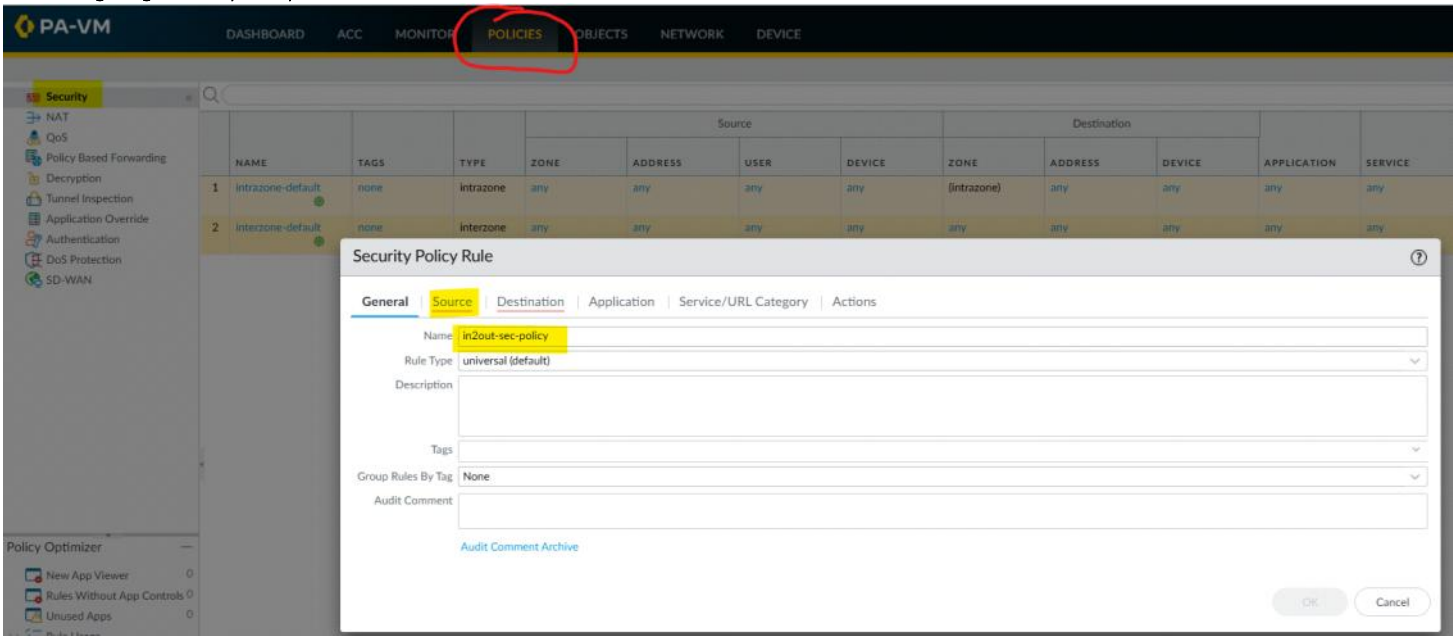


Select "Translated Packet":

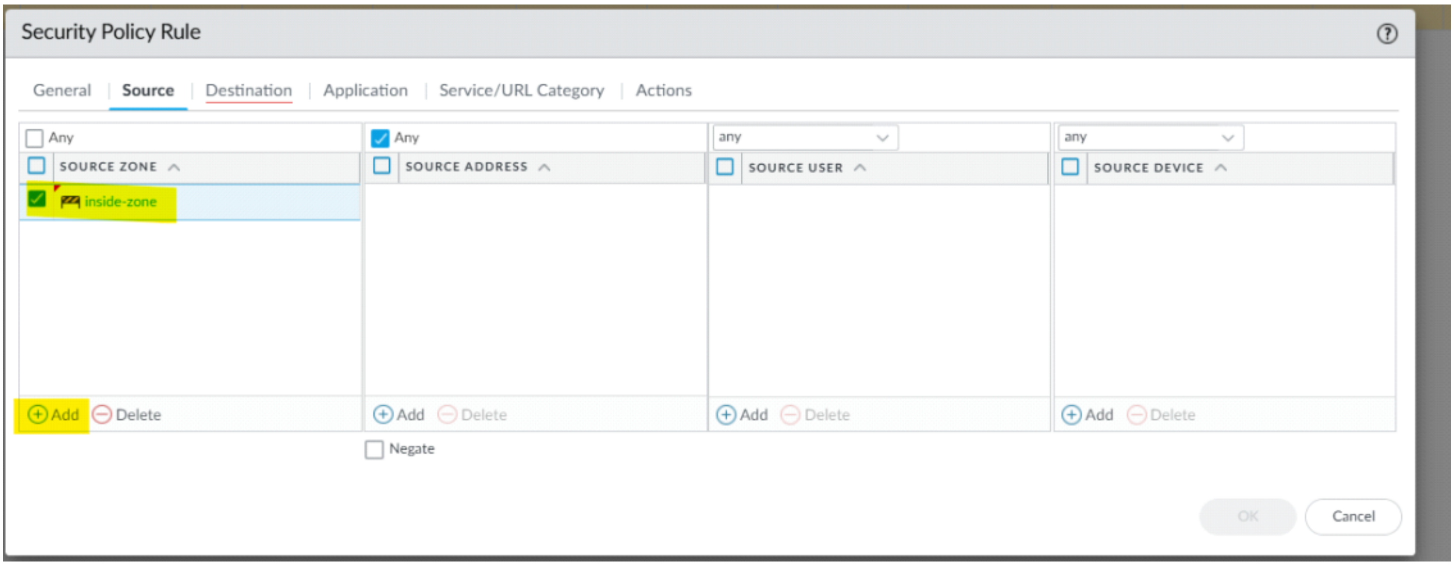




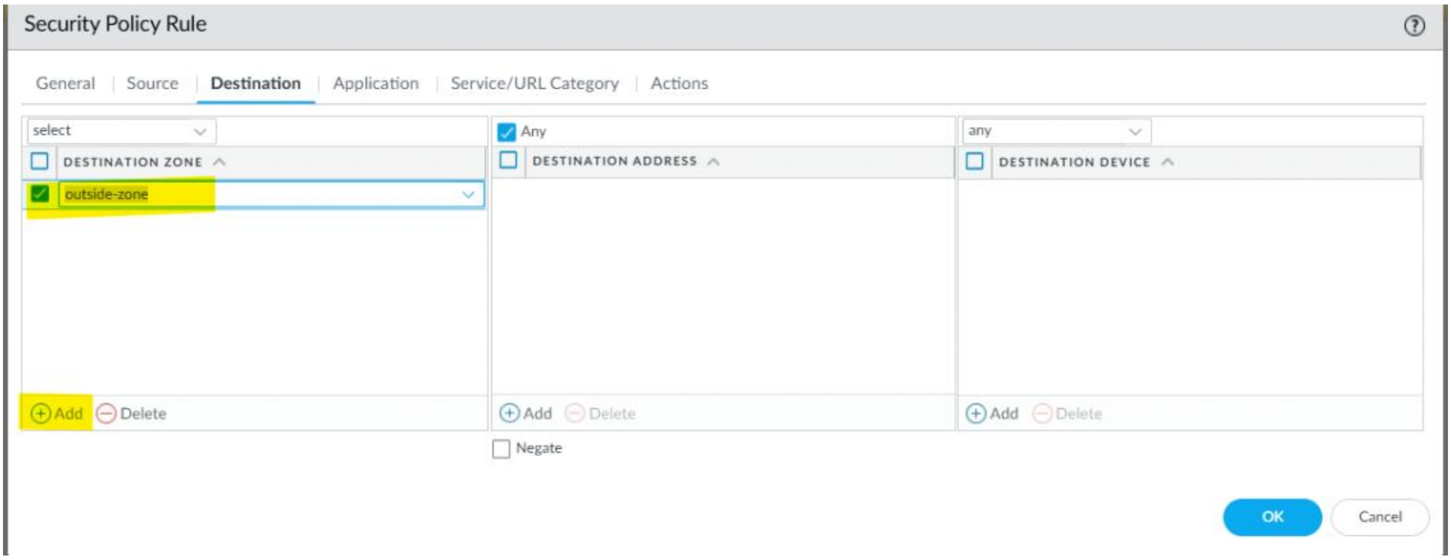
Now configuring "Security Policy":



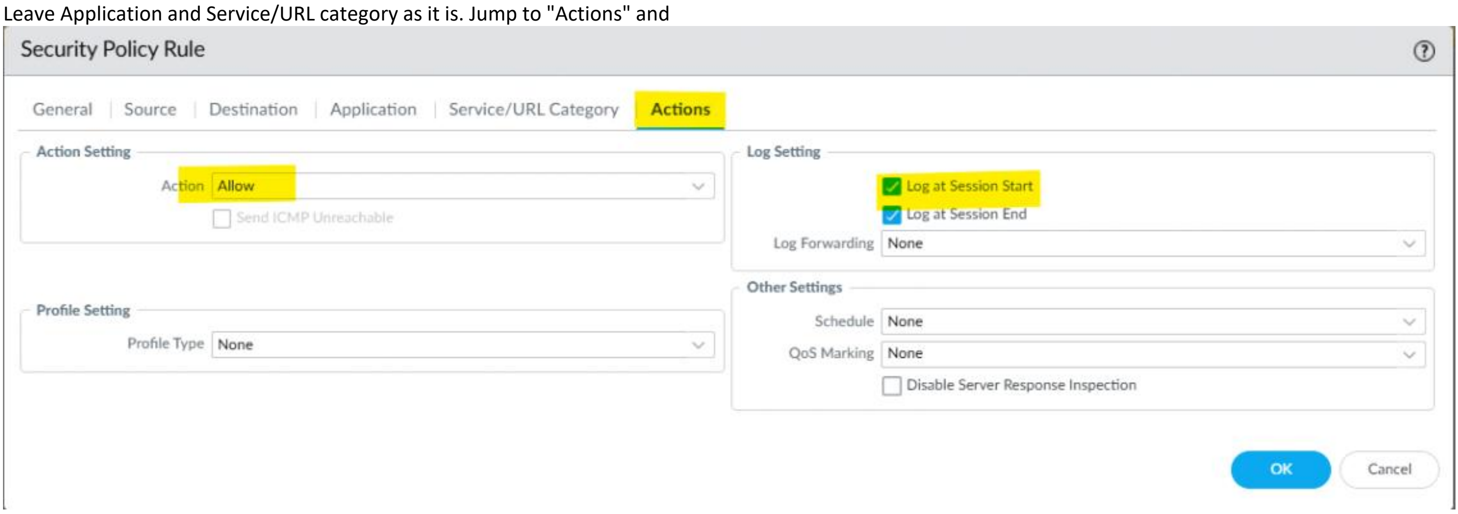
Select inside-zone for source:



Go to destination:



Leave Application and Service/URL category as it is. Jump to "Actions" and



And validate:

	NAME	TAGS	TYPE	ZONE	ADDRESS	USER	DEVICE	ZONE	ADDRESS
1	in2out-sec-policy	none	universal	inside-zone	any	any	any	outside-zone	any
2	intrazone-default	none	intrazone	any	any	any	any	(intrazone)	any
3	interzone-default	none	interzone	any	any	any	any	any	any

COMMIT the changes and you are done.

To validate the network is configured from PA-FW properly:

 # ping source 10.10.10.1 host 8.8.8.8

- This must ping, meaning everything is good.